Supplementary Material for Linear Ordering Problem: Time for a Change

March 2026

In this document we provide the following data.

- Tables 1–3 summarize the results for the single-solution setting across the considered instance sets (`rxr`, `pxp`, `os300`). The symbols ▲ and ▽ indicate that the algorithm performs statistically better or worse, respectively, than its competitor, while the absence of a symbol denotes no significant difference.
- Tables 4–30 present the results for the multi-solution setting, with one table corresponding to each combination of instance set (`rxr`, `pxp`, `os300`), archive size (m), and metric (Φ, Δ_{NN}, Δ_{SP}). The symbols ▲ and ▽ indicate that the algorithm performs statistically better or worse, respectively, than its competitor, while the absence of a symbol denotes no significant difference.
- Figures 1–18 present the scatter plots, both "Φ vs. Δ_{NN}" and "Φ vs. Δ_{SP}", for each single instance, where each point corresponds to the solution set obtained by a single execution of the algorithms.

Table 1: Results for classical setting (LOP) on os300 instances

Instance	CD-RVNS			MA-EDM		
	Best	Median	IQR	Best	Median	IQR
os300_at_2022_n300	**28243679**	28175026 ▽	35489	**28243679**	28228057 ▲	11138
os300_au_2022_n300	**42091437**	41979994 ▽	28556	**42091437**	42004540 ▲	40877
os300_be_2022_n300	**26563205**	26514166 ▽	28571	**26563205**	26527532 ▲	19636
os300_bg_2022_n300	**36423853**	36355238 ▽	39376	**36423853**	36385711 ▲	36896
os300_br_2022_n300	**38581839**	38457292 ▽	67950	**38581839**	38537140 ▲	35959
os300_ca_2022_n300	**38445729**	38373506 ▽	47844	**38445729**	38403718 ▲	53109
os300_ch_2022_n300	**33018980**	32896807 ▽	38510	**33018980**	32972636 ▲	32189
os300_cn_2022_n300	**56798580**	56632832 ▽	51386	**56798580**	56695628 ▲	53058
os300_cy_2022_n300	**39300450**	39168943 ▽	71416	**39300450**	39263457 ▲	27214
os300_cz_2022_n300	**27123568**	27061229 ▽	32171	**27123568**	27093757 ▲	14081
os300_de_2022_n300	**26935500**	26857898 ▽	26037	**26935500**	26915364 ▲	18340
os300_dk_2022_n300	**35215033**	35135597 ▽	53510	**35215033**	35156434 ▲	31932
os300_ee_2022_n300	**33789498**	33687218 ▽	37548	**33789498**	33750548 ▲	28746
os300_es_2022_n300	**26643465**	26567355 ▽	31320	**26643465**	26611429 ▲	20842
os300_fi_2022_n300	**31560107**	31487729 ▽	41325	**31560107**	31512874 ▲	22698
os300_fr_2022_n300	**32127336**	32046909 ▽	44052	**32127336**	32097975 ▲	41493
os300_gb_2022_n300	**29553210**	29487485 ▽	37957	**29553210**	29522823 ▲	23192
os300_gr_2022_n300	**30984307**	30906246 ▽	29079	**30984307**	30953620 ▲	10650
os300_hr_2022_n300	**30687985**	30602148 ▽	40126	**30687985**	30646139 ▲	24215
os300_hu_2022_n300	**28960131**	28885484 ▽	41621	**28960131**	28935349 ▲	24853
os300_id_2022_n300	**64792455**	64549234 ▽	141524	**64792455**	64724542 ▲	69094
os300_ie_2022_n300	**23356772**	23299913 ▽	37344	**23356772**	23327833 ▲	15694
os300_in_2022_n300	**38414884**	38271902 ▽	68465	**38414884**	38376672 ▲	33889
os300_it_2022_n300	**27239986**	27142874 ▽	39739	**27239986**	27190512 ▲	30394
os300_jp_2022_n300	**37384329**	37267367 ▽	74377	**37384329**	37337370 ▲	23884
os300_kr_2022_n300	**34421850**	34324784 ▽	40698	**34421850**	34387583 ▲	38809
os300_lt_2022_n300	**28023381**	27947262 ▽	18683	**28023381**	27986386 ▲	29352
os300_lu_2022_n300	**31888525**	31844253	33087	**31888525**	31841426	24743
os300_lv_2022_n300	**32935749**	32881366 ▽	29216	**32935749**	32902726 ▲	20101
os300_mt_2022_n300	**45500160**	45400004 ▽	85920	**45500160**	45438546 ▲	42388
os300_mx_2022_n300	**33047210**	32966544 ▽	54977	**33047210**	33005516 ▲	30371
os300_nl_2022_n300	**21768835**	21722606 ▽	27205	**21768835**	21760584 ▲	8528
os300_no_2022_n300	**49039776**	48860713 ▽	89084	**49039776**	48989826 ▲	59637
os300_pl_2022_n300	**23799046**	23751658 ▽	18437	**23799046**	23774682 ▲	15548
os300_pt_2022_n300	**27235539**	27161224 ▽	27593	**27235539**	27216574 ▲	21585
os300_ro_2022_n300	**37256782**	37114276 ▽	90723	**37256782**	37172353 ▲	44869
os300_ru_2022_n300	**40609705**	40528856 ▽	80311	**40609705**	40581006 ▲	16955
os300_se_2022_n300	**25872116**	25805400 ▽	27063	**25872116**	25842856 ▲	15568
os300_si_2022_n300	**31694055**	31626946 ▽	52095	**31694055**	31665518 ▲	15574
os300_sk_2022_n300	**28402484**	28314890 ▽	33192	**28402484**	28374928 ▲	26401
os300_tr_2022_n300	**34418550**	34339752 ▽	23760	**34418550**	34382844 ▲	27236
os300_tw_2022_n300	**66015071**	65878610 ▽	135368	**66015071**	65968872 ▲	50450
os300_us_2022_n300	**37101549**	36983252 ▽	68319	**37101549**	37058007 ▲	39701
os300_wa_2022_n300	**15455699**	15412038 ▽	16449	**15455699**	15433734 ▲	11498
os300_we_2022_n300	**26505810**	26469742	27542	**26505810**	26475162	21865
os300_wf_2022_n300	**32611507**	32557080	31336	**32611507**	32546784	26882
os300_wl_2022_n300	**24513154**	24447462 ▽	34520	**24513154**	24480604 ▲	13322
os300_wm_2022_n300	**16166472**	16116169 ▽	20874	**16166472**	16145740 ▲	13724
os300_za_2022_n300	**48803579**	48650110 ▽	91252	**48803579**	48764856 ▲	33564

Table 2: Results for classical setting (LOP) on pxp instances

Instance	CD-RVNS			MA-EDM		
	Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	**4997262**	4997262 ▲	0	**4997262**	4997252 ▽	68
pxp_au_2022_n155	**6509976**	6509882	94	**6509976**	6509970	74
pxp_be_2022_n155	**4840309**	4840309	0	**4840309**	4840309	66
pxp_bg_2022_n135	**5550530**	5550359	128	**5550530**	5550478	186
pxp_br_2022_n155	**5993534**	5993056 ▽	230	**5993534**	5993152 ▲	317
pxp_ca_2022_n158	**5607426**	5607426	0	**5607426**	5607426	0
pxp_ch_2022_n141	**4873413**	4873413	0	**4873413**	4873413	0
pxp_cn_2022_n169	**7846651**	7846651 ▲	0	**7846651**	7846056 ▽	861
pxp_cy_2022_n116	**3473529**	3473529 ▲	0	**3473529**	3473496 ▽	131
pxp_cz_2022_n162	**6076090**	6076090	0	**6076090**	6076090	0
pxp_de_2022_n165	**5833705**	5833280 ▽	0	**5833705**	5833705 ▲	4
pxp_dk_2022_n141	**4795237**	4795237 ▲	0	**4795237**	4795235 ▽	2
pxp_ee_2022_n126	**4271813**	4271813	0	**4271813**	4271813	0
pxp_es_2022_n162	**5505970**	5505970	0	**5505970**	5505970	61
pxp_fi_2022_n153	**5823310**	5823310	1	**5823310**	5823310	0
pxp_fr_2022_n164	**6191534**	6191534 ▲	31	**6191534**	6191503 ▽	36
pxp_gb_2022_n159	**5322836**	5322836 ▲	0	**5322836**	5322692 ▽	50
pxp_gr_2022_n130	**3400458**	3400458	0	**3400458**	3400458	0
pxp_hr_2022_n140	**4686061**	4686061	0	**4686061**	4686061	0
pxp_hu_2022_n160	**5739389**	5739235 ▽	4	**5739389**	5739389 ▲	3
pxp_id_2022_n132	**5249010**	5249010 ▲	0	**5249010**	5248495 ▽	528
pxp_ie_2022_n136	**2351111**	2351111	0	**2351111**	2351111	0
pxp_in_2022_n145	**5026649**	5026649 ▲	0	**5026649**	5026629 ▽	5
pxp_it_2022_n165	**5986644**	5986055 ▲	303	**5986644**	5986020 ▽	174
pxp_jp_2022_n159	**6439663**	6439663 ▲	0	**6439663**	6439659 ▽	4
pxp_kr_2022_n149	**5254433**	5254433 ▲	0	**5254433**	5253929 ▽	470
pxp_lt_2022_n133	**3695421**	3695421	0	**3695421**	3695421	0
pxp_lu_2022_n120	**2534779**	2534779 ▲	0	**2534779**	2534748 ▽	78
pxp_lv_2022_n132	**4580756**	4580756	0	**4580756**	4580756	0
pxp_mt_2022_n89	**2345841**	2345841	0	**2345841**	2345841	0
pxp_mx_2022_n139	**4176737**	4176712 ▲	270	**4176737**	4176446 ▽	0
pxp_nl_2022_n158	**4248155**	4248155	0	**4248155**	4248155	76
pxp_no_2022_n145	**6631865**	6631865 ▲	0	**6631865**	6631827 ▽	13
pxp_pl_2022_n166	**5465401**	5465401	0	**5465401**	5465401	24
pxp_pt_2022_n149	**4974639**	4974639 ▲	0	**4974639**	4973856 ▽	585
pxp_ro_2022_n151	**5449486**	5449486 ▲	0	**5449486**	5448906 ▽	631
pxp_ru_2022_n136	**5230190**	5230190 ▲	0	**5230190**	5230056 ▽	724
pxp_se_2022_n156	**5079306**	5079262 ▽	44	**5079306**	5079306 ▲	0
pxp_si_2022_n129	**4262916**	4262916	0	**4262916**	4262916	0
pxp_sk_2022_n151	**5016909**	5016909	0	**5016909**	5016909	746
pxp_tr_2022_n145	**5365578**	5365578 ▲	0	**5365578**	5365433 ▽	177
pxp_tw_2022_n138	**5532757**	5532757 ▲	0	**5532757**	5532556 ▽	156
pxp_us_2022_n170	**6613891**	6613891	0	**6613891**	6613891	0
pxp_wa_2022_n176	**4011427**	4011427	0	**4011427**	4011427	0
pxp_we_2022_n144	**4492970**	4492970 ▲	0	**4492970**	4492856 ▽	28
pxp_wf_2022_n141	**4829905**	4829905	0	**4829905**	4829905	0
pxp_wl_2022_n164	**4966516**	4966516 ▲	0	**4966516**	4966480 ▽	28
pxp_wm_2022_n161	**3485302**	3485302 ▲	0	**3485302**	3485292 ▽	47
pxp_za_2022_n134	**5634755**	5634755 ▲	0	**5634755**	5634406 ▽	799

Table 3: Results for classical setting (LOP) on rxr instances

Instance	**CD-RVNS**			**MA-EDM**		
	Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
rxr__1995__n49	**305741**	305741	0	**305741**	305741	0
rxr__1996__n49	**321479**	321479	0	**321479**	321479	0
rxr__1997__n49	**344182**	344182	0	**344182**	344182	0
rxr__1998__n49	**341390**	341390	0	**341390**	341390	0
rxr__1999__n49	**339587**	339587	0	**339587**	339587	0
rxr__2000__n49	**366789**	366789	0	**366789**	366789	0
rxr__2001__n49	**366155**	366155	0	**366155**	366155	0
rxr__2002__n49	**359952**	359952	0	**359952**	359952	0
rxr__2003__n49	**342326**	342326	0	**342326**	342326	0
rxr__2004__n49	**361744**	361744	0	**361744**	361744	0
rxr__2005__n49	**374491**	374491	0	**374491**	374491	0
rxr__2006__n49	**389927**	389927	0	**389927**	389927	0
rxr__2007__n49	**384053**	384053	0	**384053**	384053	0
rxr__2008__n49	**393668**	393668	0	**393668**	393668	0
rxr__2009__n49	**353109**	353109	0	**353109**	353109	0
rxr__2010__n49	**409621**	409621	0	**409621**	409621	0
rxr__2011__n49	**427225**	427225	0	**427225**	427225	0
rxr__2012__n49	**444306**	444306	0	**444306**	444306	0
rxr__2013__n49	**441753**	441753	0	**441753**	441753	0
rxr__2014__n49	**439250**	439250	0	**439250**	439250	0
rxr__2015__n49	**443549**	443549	0	**443549**	443549	0
rxr__2016__n49	**445805**	445805	0	**445805**	445805	0
rxr__2017__n49	**444129**	444129	0	**444129**	444129	0
rxr__2018__n49	**462936**	462936	0	**462936**	462936	0
rxr__2019__n49	**465593**	465593	0	**465593**	465593	0
rxr__2020__n49	**430104**	430104	0	**430104**	430104	0
rxr__2021__n49	**431939**	431939	0	**431939**	431939	0
rxr__2022__n49	**450783**	450783	0	**450783**	450783	0

Table 4: Results on MS-LOP for Φ metric and $m = 5$ on OS300 instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
os300_at_2022_n300	5	**28243679**	28175009 ▽	35495	**28243679**	28228047 ▲	11175
os300_au_2022_n300	5	**42091432**	41979994 ▽	28543	41983708	42004520 ▲	40881
os300_be_2022_n300	5	**26563201**	26514132 ▽	28558	26510456	26527437 ▲	19639
os300_bg_2022_n300	5	**36423843**	36355169 ▽	39284	36339035	36385643 ▲	36871
os300_br_2022_n300	5	**38581789**	38457272 ▽	67681	38526322	38537049 ▲	35938
os300_ca_2022_n300	5	**38445653**	38373468 ▽	47869	38359566	38403662 ▲	53095
os300_ch_2022_n300	5	**33018970**	32896789 ▽	38505	32983236	32972611 ▲	32160
os300_cn_2022_n300	5	**56798511**	56632832 ▽	51375	56694397	56695604 ▲	53063
os300_cy_2022_n300	5	**39300444**	39168914 ▽	71371	39267694	39263420 ▲	27243
os300_cz_2022_n300	5	**27123566**	27061214 ▽	32316	27112412	27093747 ▲	14053
os300_de_2022_n300	5	**26935421**	26857884 ▽	26035	26914138	26915310 ▲	18137
os300_dk_2022_n300	5	**35214798**	35135586 ▽	53680	**35214798**	35156414 ▲	31950
os300_ee_2022_n300	5	**33789490**	33687205 ▽	37512	33733094	33750484 ▲	28732
os300_es_2022_n300	5	**26643438**	26567316 ▽	31334	26605372	26611385 ▲	20940
os300_fi_2022_n300	5	**31560086**	31487716 ▽	41355	31556490	31512850 ▲	22717
os300_fr_2022_n300	5	**32127313**	32046868 ▽	44061	32084725	32097949 ▲	41377
os300_gb_2022_n300	5	**29553191**	29487442 ▽	38009	29531833	29522792 ▲	23172
os300_gr_2022_n300	5	**30984288**	30906183 ▽	29104	30933491	30953564 ▲	10688
os300_hr_2022_n300	5	**30687973**	30602106 ▽	40144	30639284	30646131 ▲	24216
os300_hu_2022_n300	5	**28960052**	28885483 ▽	41613	28953326	28935330 ▲	24764
os300_id_2022_n300	5	**64792430**	64549231 ▽	141530	64554655	64724519 ▲	69090
os300_ie_2022_n300	5	**23356772**	23299912 ▽	37338	23299556	23327833 ▲	15633
os300_in_2022_n300	5	**38414857**	38271870 ▽	68456	38407153	38376665 ▲	33908
os300_it_2022_n300	5	**27239961**	27142859 ▽	39746	27168478	27190386 ▲	30410
os300_jp_2022_n300	5	**37384325**	37267271 ▽	74417	37296035	37337360 ▲	23674
os300_kr_2022_n300	5	**34421848**	34324781 ▽	40677	34372482	34387568 ▲	38778
os300_lt_2022_n300	5	**28023369**	27947241 ▽	18691	28013722	27986381 ▲	29359
os300_lu_2022_n300	5	**31888475**	31844253	33087	31861759	31841391	24740
os300_lv_2022_n300	5	**32935645**	32881354 ▽	29197	32885892	32902640 ▲	20084
os300_mt_2022_n300	5	**45499608**	45399991 ▽	85833	45459203	45438495 ▲	42464
os300_mx_2022_n300	5	**33047210**	32966532 ▽	54971	32988830	33005509 ▲	30254
os300_nl_2022_n300	5	**21768812**	21722590 ▽	27216	21764404	21760552 ▲	8540
os300_no_2022_n300	5	**49039774**	48860684 ▽	89081	48972884	48989808 ▲	59695
os300_pl_2022_n300	5	**23799044**	23751637 ▽	18469	23768624	23774650 ▲	15544
os300_pt_2022_n300	5	**27235473**	27161209 ▽	27551	27218277	27216548 ▲	21632
os300_ro_2022_n300	5	**37256768**	37114250 ▽	90662	37243798	37172277 ▲	44920
os300_ru_2022_n300	5	**40609696**	40528844 ▽	80282	40597201	40580958 ▲	16947
os300_se_2022_n300	5	**25872051**	25805342 ▽	27091	25847481	25842839 ▲	15522
os300_si_2022_n300	5	**31694025**	31626939 ▽	52106	31675855	31665467 ▲	15557
os300_sk_2022_n300	5	**28402424**	28314846 ▽	33300	28388013	28374912 ▲	26195
os300_tr_2022_n300	5	**34418462**	34339742 ▽	23768	34372054	34382724 ▲	27280
os300_tw_2022_n300	5	**66015040**	65878608 ▽	135410	66012988	65968731 ▲	50433
os300_us_2022_n300	5	**37101531**	36983233 ▽	68358	37079504	37057946 ▲	39711
os300_wa_2022_n300	5	**15455675**	15412027 ▽	16444	15438206	15433716 ▲	11520
os300_we_2022_n300	5	**26505803**	26469735	27523	26479942	26475099	21857
os300_wf_2022_n300	5	**32611502**	32557077	31331	32512261	32546755	26873
os300_wl_2022_n300	5	**24513152**	24447453 ▽	34528	24465187	24480600 ▲	13322
os300_wm_2022_n300	5	**16166462**	16116063 ▽	20879	16145668	16145730 ▲	13706
os300_za_2022_n300	5	**48803453**	48650051 ▽	91246	48793050	48764776 ▲	33780

Table 5: Results on MS-LOP for Φ metric and $m = 10$ on OS300 instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
os300_at_2022_n300	10	**28243679**	28174989 ▽	35525	28243674	28228032 ▲	11177
os300_au_2022_n300	10	**42091432**	41979968 ▽	28532	41983638	42004503 ▲	40877
os300_be_2022_n300	10	**26563201**	26514103 ▽	28544	26510450	26527346 ▲	19672
os300_bg_2022_n300	10	**36423843**	36355112 ▽	39226	36338895	36385543 ▲	36877
os300_br_2022_n300	10	**38581789**	38457241 ▽	67602	38526271	38536952 ▲	35918
os300_ca_2022_n300	10	**38445653**	38373434 ▽	47902	38359558	38403625 ▲	53089
os300_ch_2022_n300	10	**33018970**	32896752 ▽	38471	32983198	32972577 ▲	32156
os300_cn_2022_n300	10	**56798511**	56632830 ▽	51332	56694356	56695563 ▲	53070
os300_cy_2022_n300	10	**39300444**	39168898 ▽	71333	39267689	39263384 ▲	27270
os300_cz_2022_n300	10	**27123566**	27061191 ▽	32385	27112337	27093731 ▲	14037
os300_de_2022_n300	10	**26935421**	26857869 ▽	26036	26913998	26915195 ▲	17853
os300_dk_2022_n300	10	**35214798**	35135568 ▽	53834	35214707	35156389 ▲	31986
os300_ee_2022_n300	10	**33789490**	33687189 ▽	37459	33732973	33750341 ▲	28766
os300_es_2022_n300	10	**26643438**	26567264 ▽	31350	26605354	26611338 ▲	20972
os300_fi_2022_n300	10	**31560086**	31487692 ▽	41380	31556462	31512796 ▲	22736
os300_fr_2022_n300	10	**32127313**	32046808 ▽	44037	32084676	32097909 ▲	41367
os300_gb_2022_n300	10	**29553191**	29487394 ▽	38044	29531802	29522725 ▲	23164
os300_gr_2022_n300	10	**30984288**	30906116 ▽	29169	30933402	30953508 ▲	10761
os300_hr_2022_n300	10	**30687973**	30602076 ▽	40149	30639264	30646105 ▲	24224
os300_hu_2022_n300	10	**28960052**	28885468 ▽	41564	28953271	28935306 ▲	24698
os300_id_2022_n300	10	**64792430**	64549220 ▽	141599	64554647	64724485 ▲	69051
os300_ie_2022_n300	10	**23356772**	23299906 ▽	37336	23299554	23327829 ▲	15582
os300_in_2022_n300	10	**38414857**	38271840 ▽	68422	38407139	38376652 ▲	33916
os300_it_2022_n300	10	**27239961**	27142834 ▽	39775	27168239	27190293 ▲	30422
os300_jp_2022_n300	10	**37384325**	37267212 ▽	74427	37295866	37337350 ▲	23574
os300_kr_2022_n300	10	**34421848**	34324773 ▽	40623	34372321	34387548 ▲	38750
os300_lt_2022_n300	10	**28023369**	27947205 ▽	18690	28013684	27986305 ▲	29361
os300_lu_2022_n300	10	**31888475**	31844244	33082	31861727	31841349	24735
os300_lv_2022_n300	10	**32935645**	32881321 ▽	29192	32885843	32902536 ▲	20091
os300_mt_2022_n300	10	**45499608**	45399964 ▽	85730	45459192	45438456 ▲	42441
os300_mx_2022_n300	10	**33047210**	32966516 ▽	54939	32988717	33005496 ▲	30180
os300_nl_2022_n300	10	**21768812**	21722538 ▽	27222	21764363	21760517 ▲	8589
os300_no_2022_n300	10	**49039774**	48860623 ▽	89174	48972858	48989780 ▲	59780
os300_pl_2022_n300	10	**23799044**	23751618 ▽	18497	23768501	23774627 ▲	15534
os300_pt_2022_n300	10	**27235473**	27161169 ▽	27516	27218254	27216513 ▲	21710
os300_ro_2022_n300	10	**37256768**	37114234 ▽	90653	37243761	37172149 ▲	44916
os300_ru_2022_n300	10	**40609696**	40528829 ▽	80268	40597182	40580910 ▲	16934
os300_se_2022_n300	10	**25872051**	25805264 ▽	27108	25847450	25842816 ▲	15480
os300_si_2022_n300	10	**31694025**	31626915 ▽	52125	31675813	31665436 ▲	15561
os300_sk_2022_n300	10	**28402424**	28314811 ▽	33461	28387936	28374890 ▲	26070
os300_tr_2022_n300	10	**34418462**	34339727 ▽	23786	34372001	34382660 ▲	27407
os300_tw_2022_n300	10	**66015040**	65878567 ▽	135338	66012889	65968611 ▲	50410
os300_us_2022_n300	10	**37101531**	36983210 ▽	68392	37079487	37057910 ▲	39722
os300_wa_2022_n300	10	**15455675**	15412012 ▽	16443	15438202	15433691 ▲	11538
os300_we_2022_n300	10	**26505803**	26469713	27499	26479936	26475060	21854
os300_wf_2022_n300	10	**32611502**	32557070	31130	32512238	32546732	26862
os300_wl_2022_n300	10	**24513152**	24447443 ▽	34521	24465160	24480591 ▲	13322
os300_wm_2022_n300	10	**16166462**	16116017 ▽	20889	16145629	16145704 ▲	13692
os300_za_2022_n300	10	**48803453**	48650003 ▽	91248	48793010	48764724 ▲	33876

Table 6: Results on MS-LOP for Φ metric and $m = 15$ on OS300 instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
os300_at_2022_n300	15	**28243679**	28174972 ▽	35538	28243659	28228019 ▲	11173
os300_au_2022_n300	15	**42091432**	41979940 ▽	28507	41983596	42004490 ▲	40874
os300_be_2022_n300	15	**26563201**	26514077 ▽	28534	26510443	26527274 ▲	19705
os300_bg_2022_n300	15	**36423843**	36355056 ▽	39182	36338785	36385253 ▲	36892
os300_br_2022_n300	15	**38581789**	38457204 ▽	67566	38526238	38536887 ▲	35920
os300_ca_2022_n300	15	**38445653**	38373406 ▽	47924	38359550	38403592 ▲	53105
os300_ch_2022_n300	15	**33018970**	32896717 ▽	38455	32983179	32972554 ▲	32156
os300_cn_2022_n300	15	**56798511**	56632829 ▽	51291	56694328	56695527 ▲	53086
os300_cy_2022_n300	15	**39300444**	39168886 ▽	71321	39267681	39263325 ▲	27362
os300_cz_2022_n300	15	**27123566**	27061167 ▽	32422	27112280	27093715 ▲	14032
os300_de_2022_n300	15	**26935421**	26857857 ▽	26026	26913912	26915120 ▲	17684
os300_dk_2022_n300	15	**35214798**	35135554 ▽	53997	35214620	35156368 ▲	32002
os300_ee_2022_n300	15	**33789490**	33687171 ▽	37464	33732889	33750240 ▲	28800
os300_es_2022_n300	15	**26643438**	26567219 ▽	31356	26605334	26611311 ▲	21010
os300_fi_2022_n300	15	**31560086**	31487672 ▽	41395	31556438	31512758 ▲	22754
os300_fr_2022_n300	15	**32127313**	32046758 ▽	44006	32084611	32097870 ▲	41368
os300_gb_2022_n300	15	**29553191**	29487360 ▽	38061	29531771	29522664 ▲	23152
os300_gr_2022_n300	15	**30984288**	30906053 ▽	29225	30933339	30953466 ▲	10798
os300_hr_2022_n300	15	**30687973**	30602043 ▽	40185	30639242	30646076 ▲	24217
os300_hu_2022_n300	15	**28960052**	28885450 ▽	41536	28953226	28935284 ▲	24649
os300_id_2022_n300	15	**64792430**	64549188 ▽	141631	64554636	64724458 ▲	69016
os300_ie_2022_n300	15	**23356772**	23299886 ▽	37337	23299545	23327825 ▲	15550
os300_in_2022_n300	15	**38414857**	38271806 ▽	68396	38407132	38376639 ▲	33916
os300_it_2022_n300	15	**27239961**	27142812 ▽	39782	27168027	27190239 ▲	30434
os300_jp_2022_n300	15	**37384325**	37267144 ▽	74438	37295776	37337319 ▲	23530
os300_kr_2022_n300	15	**34421848**	34324764 ▽	40596	34372042	34387527 ▲	38735
os300_lt_2022_n300	15	**28023369**	27947180 ▽	18696	28013646	27986229 ▲	29362
os300_lu_2022_n300	15	**31888475**	31844231	33081	31861684	31841316	24727
os300_lv_2022_n300	15	**32935645**	32881278 ▽	29182	32885807	32902428 ▲	20090
os300_mt_2022_n300	15	**45499608**	45399925 ▽	85637	45459180	45438424 ▲	42413
os300_mx_2022_n300	15	**33047210**	32966504 ▽	54922	32988630	33005480 ▲	30149
os300_nl_2022_n300	15	**21768812**	21722492 ▽	27220	21764337	21760483 ▲	8637
os300_no_2022_n300	15	**49039774**	48860560 ▽	89226	48972843	48989759 ▲	59863
os300_pl_2022_n300	15	**23799044**	23751604 ▽	18515	23768426	23774607 ▲	15536
os300_pt_2022_n300	15	**27235473**	27161135 ▽	27504	27218232	27216486 ▲	21738
os300_ro_2022_n300	15	**37256768**	37114222 ▽	90661	37243693	37172044 ▲	44888
os300_ru_2022_n300	15	**40609696**	40528815 ▽	80273	40597165	40580872 ▲	16912
os300_se_2022_n300	15	**25872051**	25805193 ▽	27113	25847424	25842790 ▲	15431
os300_si_2022_n300	15	**31694025**	31626893 ▽	52142	31675787	31665410 ▲	15551
os300_sk_2022_n300	15	**28402424**	28314784 ▽	33553	28387873	28374865 ▲	25971
os300_tr_2022_n300	15	**34418462**	34339715 ▽	23804	34371965	34382629 ▲	27448
os300_tw_2022_n300	15	**66015040**	65878536 ▽	135297	66012771	65968502 ▲	50377
os300_us_2022_n300	15	**37101531**	36983188 ▽	68419	37079473	37057846 ▲	39728
os300_wa_2022_n300	15	**15455675**	15411999 ▽	16441	15438197	15433674 ▲	11553
os300_we_2022_n300	15	**26505803**	26469675	27483	26479930	26475028	21850
os300_wf_2022_n300	15	**32611502**	32557061	31039	32512228	32546715	26834
os300_wl_2022_n300	15	**24513152**	24447434 ▽	34509	24465136	24480576 ▲	13325
os300_wm_2022_n300	15	**16166462**	16115995 ▽	20896	16145600	16145683 ▲	13690
os300_za_2022_n300	15	**48803453**	48649948 ▽	91222	48792989	48764678 ▲	33912

Table 7: Results on MS-LOP for Φ metric and $m = 5$ on PXP instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
pxp_at_2022_n155	5	**4997262**	4997262 ▲	0	**4997262**	4997252 ▽	68
pxp_au_2022_n155	5	**6509976**	6509882	94	6509967	6509970	74
pxp_be_2022_n155	5	**4840309**	4840309	0	**4840309**	4840309	66
pxp_bg_2022_n135	5	**5550530**	5550359	128	5550478	5550478	186
pxp_br_2022_n155	5	**5993534**	5993056 ▽	230	5993099	5993152 ▲	317
pxp_ca_2022_n158	5	**5607426**	5607426	0	**5607426**	5607426	0
pxp_ch_2022_n141	5	**4873413**	4873413	0	**4873413**	4873413	0
pxp_cn_2022_n169	5	**7846651**	7846651 ▲	0	7846389	7846056 ▽	861
pxp_cy_2022_n116	5	**3473529**	3473529 ▲	0	3473386	3473496 ▽	131
pxp_cz_2022_n162	5	**6076090**	6076090	0	**6076090**	6076090	0
pxp_de_2022_n165	5	**5833705**	5833280 ▽	0	**5833705**	5833705 ▲	4
pxp_dk_2022_n141	5	**4795237**	4795237 ▲	0	4795235	4795235 ▽	2
pxp_ee_2022_n126	5	**4271813**	4271813	0	**4271813**	4271813	0
pxp_es_2022_n162	5	**5505970**	5505970	0	5505909	5505970	61
pxp_fi_2022_n153	5	**5823310**	5823310	1	5823309	5823310	0
pxp_fr_2022_n164	5	**6191534**	6191534 ▲	31	6191529	6191503 ▽	36
pxp_gb_2022_n159	5	**5322836**	5322836 ▲	0	5322742	5322692 ▽	50
pxp_gr_2022_n130	5	**3400458**	3400458	0	**3400458**	3400458	0
pxp_hr_2022_n140	5	**4686061**	4686061	0	**4686061**	4686061	0
pxp_hu_2022_n160	5	**5739389**	5739235 ▽	4	**5739389**	5739389 ▲	3
pxp_id_2022_n132	5	**5249010**	5249010 ▲	0	5248855	5248495 ▽	528
pxp_ie_2022_n136	5	**2351111**	2351111	0	**2351111**	2351111	0
pxp_in_2022_n145	5	**5026649**	5026649 ▲	0	5026629	5026629 ▽	5
pxp_it_2022_n165	5	**5986644**	5986055 ▲	303	5986037	5986020 ▽	174
pxp_jp_2022_n159	5	**6439663**	6439663 ▲	0	6439659	6439659 ▽	4
pxp_kr_2022_n149	5	**5254433**	5254433 ▲	0	**5254433**	5253929 ▽	470
pxp_lt_2022_n133	5	**3695421**	3695421	0	**3695421**	3695421	0
pxp_lu_2022_n120	5	**2534779**	2534779 ▲	0	**2534779**	2534748 ▽	78
pxp_lv_2022_n132	5	**4580756**	4580756	0	**4580756**	4580756	0
pxp_mt_2022_n89	5	**2345841**	2345841	0	**2345841**	2345841	0
pxp_mx_2022_n139	5	**4176737**	4176712 ▲	270	4176446	4176446 ▽	0
pxp_nl_2022_n158	5	**4248155**	4248155	0	**4248155**	4248155	76
pxp_no_2022_n145	5	**6631865**	6631865 ▲	0	**6631865**	6631827 ▽	13
pxp_pl_2022_n166	5	**5465401**	5465401	0	**5465401**	5465401	24
pxp_pt_2022_n149	5	**4974639**	4974639 ▲	0	4974624	4973856 ▽	585
pxp_ro_2022_n151	5	**5449486**	5449486 ▲	0	5448898	5448906 ▽	631
pxp_ru_2022_n136	5	**5230190**	5230190 ▲	0	5229836	5230056 ▽	724
pxp_se_2022_n156	5	**5079306**	5079262 ▽	44	**5079306**	5079306 ▲	0
pxp_si_2022_n129	5	**4262916**	4262916	0	**4262916**	4262916	0
pxp_sk_2022_n151	5	**5016909**	5016909	0	**5016909**	5016909	746
pxp_tr_2022_n145	5	**5365578**	5365578 ▲	0	5365349	5365433 ▽	177
pxp_tw_2022_n138	5	**5532757**	5532757 ▲	0	5532672	5532556 ▽	156
pxp_us_2022_n170	5	**6613891**	6613891	0	**6613891**	6613891	0
pxp_wa_2022_n176	5	**4011427**	4011427	0	**4011427**	4011427	0
pxp_we_2022_n144	5	**4492970**	4492970 ▲	0	4492856	4492856 ▽	28
pxp_wf_2022_n141	5	**4829905**	4829905	0	**4829905**	4829905	0
pxp_wl_2022_n164	5	**4966516**	4966516 ▲	0	4966480	4966480 ▽	28
pxp_wm_2022_n161	5	**3485302**	3485302 ▲	0	3485251	3485292 ▽	47
pxp_za_2022_n134	5	**5634755**	5634755 ▲	0	**5634755**	5634406 ▽	799

Table 8: Results on MS-LOP for Φ metric and $m = 10$ on PXP instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
pxp_at_2022_n155	10	**4997262**	4997262 ▲	0	**4997262**	4997252 ▽	68
pxp_au_2022_n155	10	**6509976**	6509882	94	6509967	6509970	74
pxp_be_2022_n155	10	**4840309**	4840309	0	**4840309**	4840309	66
pxp_bg_2022_n135	10	**5550530**	5550359	128	5550478	5550478	186
pxp_br_2022_n155	10	**5993534**	5993056 ▽	230	5993099	5993152 ▲	317
pxp_ca_2022_n158	10	**5607426**	5607426	0	**5607426**	5607426	0
pxp_ch_2022_n141	10	**4873413**	4873413	0	**4873413**	4873413	0
pxp_cn_2022_n169	10	**7846651**	7846651 ▲	0	7846389	7846056 ▽	861
pxp_cy_2022_n116	10	**3473529**	3473529 ▲	0	3473386	3473496 ▽	131
pxp_cz_2022_n162	10	**6076090**	6076090	0	**6076090**	6076090	0
pxp_de_2022_n165	10	**5833705**	5833280 ▽	0	**5833705**	5833705 ▲	4
pxp_dk_2022_n141	10	**4795237**	4795237 ▲	0	4795235	4795235 ▽	2
pxp_ee_2022_n126	10	**4271813**	4271813	0	**4271813**	4271813	0
pxp_es_2022_n162	10	**5505970**	5505970	0	5505909	5505970	61
pxp_fi_2022_n153	10	**5823310**	5823310	1	5823309	5823310	0
pxp_fr_2022_n164	10	**6191534**	6191534 ▲	31	6191529	6191503 ▽	36
pxp_gb_2022_n159	10	**5322836**	5322836 ▲	0	5322742	5322692 ▽	50
pxp_gr_2022_n130	10	**3400458**	3400458	0	**3400458**	3400458	0
pxp_hr_2022_n140	10	**4686061**	4686061	0	**4686061**	4686061	0
pxp_hu_2022_n160	10	**5739389**	5739235 ▽	4	**5739389**	5739389 ▲	3
pxp_id_2022_n132	10	**5249010**	5249010 ▲	0	5248855	5248495 ▽	528
pxp_ie_2022_n136	10	**2351111**	2351111	0	**2351111**	2351111	0
pxp_in_2022_n145	10	**5026649**	5026649 ▲	0	5026629	5026629 ▽	5
pxp_it_2022_n165	10	**5986644**	5986055 ▲	303	5986037	5986020 ▽	174
pxp_jp_2022_n159	10	**6439663**	6439663 ▲	0	6439659	6439659 ▽	4
pxp_kr_2022_n149	10	**5254433**	5254433 ▲	0	**5254433**	5253929 ▽	470
pxp_lt_2022_n133	10	**3695421**	3695421	0	**3695421**	3695421	0
pxp_lu_2022_n120	10	**2534779**	2534779 ▲	0	**2534779**	2534748 ▽	78
pxp_lv_2022_n132	10	**4580756**	4580756	0	**4580756**	4580756	0
pxp_mt_2022_n89	10	**2345841**	2345841	0	**2345841**	2345841	0
pxp_mx_2022_n139	10	**4176737**	4176712 ▲	270	4176446	4176446 ▽	0
pxp_nl_2022_n158	10	**4248155**	4248155	0	**4248155**	4248155	76
pxp_no_2022_n145	10	**6631865**	6631865 ▲	0	**6631865**	6631827 ▽	13
pxp_pl_2022_n166	10	**5465401**	5465401	0	**5465401**	5465401	24
pxp_pt_2022_n149	10	**4974639**	4974639 ▲	0	4974624	4973856 ▽	585
pxp_ro_2022_n151	10	**5449486**	5449486 ▲	0	5448898	5448906 ▽	631
pxp_ru_2022_n136	10	**5230190**	5230190 ▲	0	5229836	5230056 ▽	724
pxp_se_2022_n156	10	**5079306**	5079262 ▽	44	**5079306**	5079306 ▲	0
pxp_si_2022_n129	10	**4262916**	4262916	0	**4262916**	4262916	0
pxp_sk_2022_n151	10	**5016909**	5016909	0	**5016909**	5016909	746
pxp_tr_2022_n145	10	**5365578**	5365578 ▲	0	5365349	5365433 ▽	177
pxp_tw_2022_n138	10	**5532757**	5532757 ▲	0	5532672	5532556 ▽	156
pxp_us_2022_n170	10	**6613891**	6613891	0	**6613891**	6613891	0
pxp_wa_2022_n176	10	**4011427**	4011427	0	**4011427**	4011427	0
pxp_we_2022_n144	10	**4492970**	4492970 ▲	0	4492856	4492856 ▽	28
pxp_wf_2022_n141	10	**4829905**	4829905	0	**4829905**	4829905	0
pxp_wl_2022_n164	10	**4966516**	4966516 ▲	0	4966480	4966480 ▽	28
pxp_wm_2022_n161	10	**3485302**	3485302 ▲	0	3485251	3485292 ▽	47
pxp_za_2022_n134	10	**5634755**	5634755 ▲	0	**5634755**	5634406 ▽	799

Table 9: Results on MS-LOP for Φ metric and $m = 15$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	15	**4997262**	4997262 ▲	0	**4997262**	4997252 ▽	68
pxp_au_2022_n155	15	**6509976**	6509882	94	6509967	6509970	74
pxp_be_2022_n155	15	**4840309**	4840309	0	**4840309**	4840309	66
pxp_bg_2022_n135	15	**5550530**	5550359	128	5550478	5550478	186
pxp_br_2022_n155	15	**5993534**	5993056 ▽	230	5993099	5993152 ▲	317
pxp_ca_2022_n158	15	**5607426**	5607426	0	**5607426**	5607426	0
pxp_ch_2022_n141	15	**4873413**	4873413	0	**4873413**	4873413	0
pxp_cn_2022_n169	15	**7846651**	7846651 ▲	0	7846389	7846056 ▽	861
pxp_cy_2022_n116	15	**3473529**	3473529 ▲	0	3473386	3473496 ▽	131
pxp_cz_2022_n162	15	**6076090**	6076090	0	**6076090**	6076090	0
pxp_de_2022_n165	15	**5833705**	5833280 ▽	0	**5833705**	5833705 ▲	4
pxp_dk_2022_n141	15	**4795237**	4795237 ▲	0	4795235	4795235 ▽	2
pxp_ee_2022_n126	15	**4271813**	4271813	0	**4271813**	4271813	0
pxp_es_2022_n162	15	**5505970**	5505970	0	5505909	5505970	61
pxp_fi_2022_n153	15	**5823310**	5823310	1	5823309	5823310	0
pxp_fr_2022_n164	15	**6191534**	6191534 ▲	31	6191529	6191503 ▽	36
pxp_gb_2022_n159	15	**5322836**	5322836 ▲	0	5322742	5322692 ▽	50
pxp_gr_2022_n130	15	**3400458**	3400458	0	**3400458**	3400458	0
pxp_hr_2022_n140	15	**4686061**	4686061	0	**4686061**	4686061	0
pxp_hu_2022_n160	15	**5739389**	5739235 ▽	4	**5739389**	5739389 ▲	3
pxp_id_2022_n132	15	**5249010**	5249010 ▲	0	5248855	5248495 ▽	528
pxp_ie_2022_n136	15	**2351111**	2351111	0	**2351111**	2351111	0
pxp_in_2022_n145	15	**5026649**	5026649 ▲	0	5026629	5026629 ▽	5
pxp_it_2022_n165	15	**5986644**	5986055 ▲	303	5986037	5986020 ▽	174
pxp_jp_2022_n159	15	**6439663**	6439663 ▲	0	6439659	6439659 ▽	4
pxp_kr_2022_n149	15	**5254433**	5254433 ▲	0	**5254433**	5253929 ▽	470
pxp_lt_2022_n133	15	**3695421**	3695421	0	**3695421**	3695421	0
pxp_lu_2022_n120	15	**2534779**	2534779 ▲	0	**2534779**	2534748 ▽	78
pxp_lv_2022_n132	15	**4580756**	4580756	0	**4580756**	4580756	0
pxp_mt_2022_n89	15	**2345841**	2345841	0	**2345841**	2345841	0
pxp_mx_2022_n139	15	**4176737**	4176712 ▲	270	4176446	4176446 ▽	0
pxp_nl_2022_n158	15	**4248155**	4248155	0	**4248155**	4248155	76
pxp_no_2022_n145	15	**6631865**	6631865 ▲	0	**6631865**	6631827 ▽	13
pxp_pl_2022_n166	15	**5465401**	5465401	0	**5465401**	5465401	24
pxp_pt_2022_n149	15	**4974639**	4974639 ▲	0	4974624	4973856 ▽	585
pxp_ro_2022_n151	15	**5449486**	5449486 ▲	0	5448898	5448906 ▽	631
pxp_ru_2022_n136	15	**5230190**	5230190 ▲	0	5229836	5230056 ▽	724
pxp_se_2022_n156	15	**5079306**	5079262 ▽	44	**5079306**	5079306 ▲	0
pxp_si_2022_n129	15	**4262916**	4262916	0	**4262916**	4262916	0
pxp_sk_2022_n151	15	**5016909**	5016909	0	**5016909**	5016909	746
pxp_tr_2022_n145	15	**5365578**	5365578 ▲	0	5365349	5365433 ▽	177
pxp_tw_2022_n138	15	**5532757**	5532757 ▲	0	5532672	5532556 ▽	156
pxp_us_2022_n170	15	**6613891**	6613891	0	**6613891**	6613891	0
pxp_wa_2022_n176	15	**4011427**	4011427	0	**4011427**	4011427	0
pxp_we_2022_n144	15	**4492970**	4492970 ▲	0	4492856	4492856 ▽	28
pxp_wf_2022_n141	15	**4829905**	4829905	0	**4829905**	4829905	0
pxp_wl_2022_n164	15	**4966516**	4966516 ▲	0	4966480	4966480 ▽	28
pxp_wm_2022_n161	15	**3485302**	3485302 ▲	0	3485251	3485292 ▽	47
pxp_za_2022_n134	15	**5634755**	5634755 ▲	0	**5634755**	5634406 ▽	799

Table 10: Results on MS-LOP for Φ metric and $m = 5$ on RXR instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
rxr_1995_n49	5	**305714**	305714	0	**305714**	305714	0
rxr_1996_n49	5	**321398**	321395 ▽	0	**321398**	321397 ▲	3
rxr_1997_n49	5	**344171**	344169 ▽	7	**344171**	344171 ▲	0
rxr_1998_n49	5	**341353**	341353	0	**341353**	341353	0
rxr_1999_n49	5	**339578**	339578	0	**339578**	339578	0
rxr_2000_n49	5	**366789**	366789	0	**366789**	366789	0
rxr_2001_n49	5	**366136**	366136	0	**366136**	366136	0
rxr_2002_n49	5	**359941**	359941	0	**359941**	359941	0
rxr_2003_n49	5	**342297**	342273	0	342273	342273	0
rxr_2004_n49	5	**361730**	361730	0	**361730**	361730	0
rxr_2005_n49	5	**374472**	374472	1	**374472**	374472	0
rxr_2006_n49	5	**389869**	389861 ▽	3	**389869**	389869 ▲	0
rxr_2007_n49	5	**384025**	384010	0	384010	384010	0
rxr_2008_n49	5	**393623**	393623	0	**393623**	393623	0
rxr_2009_n49	5	**353102**	353102	0	**353102**	353102	0
rxr_2010_n49	5	**409574**	409574	0	**409574**	409574	0
rxr_2011_n49	5	**427121**	427117	5	427117	427117	0
rxr_2012_n49	5	**444297**	444246	0	444246	444246	0
rxr_2013_n49	5	**441715**	441715	0	**441715**	441715	0
rxr_2014_n49	5	**439131**	439086 ▽	5	**439131**	439131 ▲	0
rxr_2015_n49	5	**443533**	443533	0	**443533**	443533	0
rxr_2016_n49	5	**445787**	445763 ▽	6	**445787**	445787 ▲	0
rxr_2017_n49	5	**443845**	443610	28	443630	443596	108
rxr_2018_n49	5	**462885**	462885	0	**462885**	462885	0
rxr_2019_n49	5	**465456**	465258 ▽	80	**465456**	465456 ▲	0
rxr_2020_n49	5	**430099**	430099	0	**430099**	430099	0
rxr_2021_n49	5	**431894**	431880 ▲	28	431866	431866 ▽	1
rxr_2022_n49	5	**450772**	450772	0	**450772**	450772	0

Table 11: Results on MS-LOP for Φ metric and $m = 10$ on RXR instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
rxr_1995_n49	10	**305714**	305625 ▽	3	305629	305629 ▲	0
rxr_1996_n49	10	**321398**	321395	21	321375	321375	7
rxr_1997_n49	10	**344171**	344146 ▽	13	344152	344152 ▲	1
rxr_1998_n49	10	**341353**	341283 ▽	21	**341353**	341337 ▲	16
rxr_1999_n49	10	**339578**	339550 ▽	4	339557	339557 ▲	0
rxr_2000_n49	10	**366789**	366788	0	366788	366788	0
rxr_2001_n49	10	**366136**	366084	0	366084	366084	5
rxr_2002_n49	10	**359941**	359927	0	359921	359927	1
rxr_2003_n49	10	**342297**	342211 ▽	5	342253	342249 ▲	18
rxr_2004_n49	10	**361730**	361691	0	361691	361691	0
rxr_2005_n49	10	**374472**	374439	7	374439	374429	31
rxr_2006_n49	10	**389869**	389824	17	389825	389828	16
rxr_2007_n49	10	**384025**	384001 ▽	22	384001	384010 ▲	10
rxr_2008_n49	10	**393623**	393585 ▽	2	393588	393589 ▲	4
rxr_2009_n49	10	**353102**	353082 ▽	5	353085	353085 ▲	0
rxr_2010_n49	10	**409574**	409394 ▽	91	409536	409518 ▲	18
rxr_2011_n49	10	**427121**	427117 ▲	5	427082	427082 ▽	0
rxr_2012_n49	10	**444297**	444212	26	444246	444210	26
rxr_2013_n49	10	**441715**	441564	0	441588	441564	33
rxr_2014_n49	10	**439131**	438783	127	438846	438844	82
rxr_2015_n49	10	**443533**	443364	9	443274	443369	90
rxr_2016_n49	10	**445787**	445580 ▽	56	445742	445628 ▲	116
rxr_2017_n49	10	**443845**	443377 ▽	65	443630	443563 ▲	115
rxr_2018_n49	10	**462885**	462799 ▲	1	462797	462795 ▽	27
rxr_2019_n49	10	**465456**	465258 ▽	80	**465456**	465456 ▲	0
rxr_2020_n49	10	**430099**	430090 ▽	3	430094	430094 ▲	1
rxr_2021_n49	10	**431894**	431880 ▲	28	431846	431856 ▽	19
rxr_2022_n49	10	**450772**	450753	0	450744	450753	1

Table 12: Results on MS-LOP for Φ metric and $m = 15$ on RXR instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
rxr__1995__n49	15	**305714**	305573 ▽	4	305580	305580 ▲	0
rxr__1996__n49	15	**321398**	321395 ▲	21	321375	321375 ▽	7
rxr__1997__n49	15	**344171**	344130 ▽	15	344139	344138 ▲	2
rxr__1998__n49	15	**341353**	341283 ▽	21	**341353**	341337 ▲	16
rxr__1999__n49	15	**339578**	339531 ▽	3	339536	339536 ▲	1
rxr__2000__n49	15	**366789**	366787	0	366787	366787	0
rxr__2001__n49	15	**366136**	366039 ▽	0	366084	366054 ▲	17
rxr__2002__n49	15	**359941**	359900 ▲	9	359876	359892 ▽	16
rxr__2003__n49	15	**342297**	342211 ▽	5	342253	342249 ▲	18
rxr__2004__n49	15	**361730**	361658 ▽	5	361659	361660 ▲	6
rxr__2005__n49	15	**374472**	374326 ▽	14	374381	374361 ▲	34
rxr__2006__n49	15	**389869**	389824	17	389825	389828	16
rxr__2007__n49	15	**384025**	384001 ▽	22	384001	384010 ▲	10
rxr__2008__n49	15	**393623**	393544 ▽	7	393555	393553 ▲	8
rxr__2009__n49	15	**353102**	353065 ▽	14	353069	353069 ▲	0
rxr__2010__n49	15	**409574**	409394 ▽	101	409536	409518 ▲	18
rxr__2011__n49	15	**427121**	427117 ▲	5	427082	427082 ▽	0
rxr__2012__n49	15	**444297**	444212	26	444246	444210	26
rxr__2013__n49	15	**441715**	441387 ▽	114	441588	441560 ▲	52
rxr__2014__n49	15	**439131**	438783	127	438773	438762	84
rxr__2015__n49	15	**443533**	443280 ▽	132	443274	443369 ▲	152
rxr__2016__n49	15	**445787**	445580 ▲	53	445573	445494 ▽	77
rxr__2017__n49	15	**443845**	443373 ▽	64	443630	443563 ▲	115
rxr__2018__n49	15	**462885**	462692 ▽	23	462740	462736 ▲	42
rxr__2019__n49	15	**465456**	465258 ▽	80	**465456**	465456 ▲	0
rxr__2020__n49	15	**430099**	430081 ▽	3	430088	430088 ▲	2
rxr__2021__n49	15	**431894**	431880 ▲	28	431846	431856 ▽	19
rxr__2022__n49	15	**450772**	450698 ▽	16	450744	450738 ▲	13

Table 13: Results on MS-LOP for Δ_{NN} metric and $m = 5$ on OS300 instances

Instance	**m**	MS-CD-RVNS			MS-MA-EDM		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
os300_at_2022_n300	5	**8**	25.0	34.75	7	17.0	15.75
os300_au_2022_n300	5	10	13.0	32.00	**31**	12.5	21.50
os300_be_2022_n300	5	**11**	14.5	15.50	**11**	27.0	27.00
os300_bg_2022_n300	5	30	32.0	28.00	**42**	30.5	41.00
os300_br_2022_n300	5	17	21.5	31.00	**34**	16.0	13.75
os300_ca_2022_n300	5	**72**	13.0	35.25	8	20.5	26.00
os300_ch_2022_n300	5	10	21.0	20.25	**54**	21.5	26.50
os300_cn_2022_n300	5	**21**	14.5	14.75	14	18.0	20.75
os300_cy_2022_n300	5	**24**	22.0	9.75	6	18.5	30.75
os300_cz_2022_n300	5	48	21.0	31.25	**68**	28.5	33.50
os300_de_2022_n300	5	7	25.0	30.50	**37**	26.5	25.75
os300_dk_2022_n300	5	15	26.0	39.50	**50**	19.5	22.00
os300_ee_2022_n300	5	10	18.0	27.75	**26**	32.0	21.50
os300_es_2022_n300	5	9	24.0	20.75	**15**	18.0	22.75
os300_fi_2022_n300	5	6	23.5	18.50	**85**	21.5	24.25
os300_fr_2022_n300	5	8	31.5	29.75	**14**	30.0	23.00
os300_gb_2022_n300	5	70	24.5	30.25	**80**	27.5	39.00
os300_gr_2022_n300	5	**125**	13.0	15.50	38	19.0	28.75
os300_hr_2022_n300	5	22	20.5	29.25	**25**	23.5	15.25
os300_hu_2022_n300	5	**16**	20.0	19.75	9	23.0	21.50
os300_id_2022_n300	5	18	20.0	22.00	**24**	26.0	23.25
os300_ie_2022_n300	5	**7**	9.5	17.75	6	12.0	13.75
os300_in_2022_n300	5	5	19.0	16.00	**10**	13.0	11.75
os300_it_2022_n300	5	**75**	28.5	37.75	21	21.0	29.00
os300_jp_2022_n300	5	37	20.5	25.25	**43**	27.0	30.75
os300_kr_2022_n300	5	**79**	13.0	19.50	9	13.5	30.75
os300_lt_2022_n300	5	47	21.0	34.75	**165**	28.5	35.25
os300_lu_2022_n300	5	34	10.5	26.00	**101**	17.5	22.00
os300_lv_2022_n300	5	**59**	27.0	22.25	46	26.5	23.75
os300_mt_2022_n300	5	**127**	28.0	29.75	11	14.5	10.25
os300_mx_2022_n300	5	15	16.0	18.50	**80**	20.0	19.75
os300_nl_2022_n300	5	**32**	19.0	17.75	18	22.5	26.50
os300_no_2022_n300	5	11	17.5	23.00	**32**	19.0	21.50
os300_pl_2022_n300	5	**17**	21.5	19.00	15	27.5	30.50
os300_pt_2022_n300	5	5	20.5	22.50	**26**	23.5	16.50
os300_ro_2022_n300	5	18	17.0	23.25	**19**	23.0	26.50
os300_ru_2022_n300	5	**21**	21.5	31.00	8	22.5	25.75
os300_se_2022_n300	5	**55**	33.0	42.25	16	28.5	23.00
os300_si_2022_n300	5	51	29.0	39.25	**89**	25.0	35.50
os300_sk_2022_n300	5	8	32.0	30.50	**34**	21.0	27.75
os300_tr_2022_n300	5	7	28.5	37.00	**13**	22.0	33.25
os300_tw_2022_n300	5	**13**	21.5	17.50	5	24.5	32.00
os300_us_2022_n300	5	**33**	17.5	14.75	8	18.0	15.25
os300_wa_2022_n300	5	**59**	23.0	46.75	33	24.0	20.25
os300_we_2022_n300	5	30	22.0	27.00	**39**	26.0	19.50
os300_wf_2022_n300	5	23	36.0	30.50	**58**	27.5	29.00
os300_wl_2022_n300	5	11	19.0	26.50	**93**	18.5	24.50
os300_wm_2022_n300	5	**9**	29.0	40.75	**9**	14.5	25.75
os300_za_2022_n300	5	**10**	14.5	27.25	5	19.5	25.50

Table 14: Results on MS-LOP for Δ_{NN} metric and $m = 10$ on OS300 instances

Instance	**m**	MS-CD-RVNS			MS-MA-EDM		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
os300_at_2022_n300	10	**26**	47.5 ▲	50.50	14	23.0 ▽	30.50
os300_au_2022_n300	10	10	34.0	52.50	**104**	21.0	49.25
os300_be_2022_n300	10	**56**	32.5	45.50	12	45.0	36.50
os300_bg_2022_n300	10	53	69.5 ▲	45.50	**75**	47.0 ▽	47.00
os300_br_2022_n300	10	**87**	53.5 ▲	49.75	69	32.0 ▽	37.75
os300_ca_2022_n300	10	**70**	50.5	56.75	12	25.5	40.50
os300_ch_2022_n300	10	15	44.0	35.25	**55**	49.5	39.75
os300_cn_2022_n300	10	**102**	42.5	52.25	19	28.5	31.25
os300_cy_2022_n300	10	**157**	59.5 ▲	90.00	14	31.5 ▽	51.25
os300_cz_2022_n300	10	51	51.0	44.00	**53**	41.0	39.75
os300_de_2022_n300	10	19	44.5	52.75	**78**	41.0	44.00
os300_dk_2022_n300	10	28	44.0	65.25	**77**	33.0	43.00
os300_ee_2022_n300	10	13	48.5	55.50	**41**	53.0	34.75
os300_es_2022_n300	10	**21**	47.0 ▲	43.25	10	27.0 ▽	29.50
os300_fi_2022_n300	10	13	56.5	57.25	**69**	38.5	49.25
os300_fr_2022_n300	10	**62**	70.5	42.25	29	42.0	40.75
os300_gb_2022_n300	10	77	59.5	70.00	**88**	43.0	55.75
os300_gr_2022_n300	10	**162**	31.5	51.50	144	44.0	47.00
os300_hr_2022_n300	10	44	53.0	43.50	**50**	31.5	34.00
os300_hu_2022_n300	10	**30**	43.0	38.75	29	42.0	36.00
os300_id_2022_n300	10	**46**	59.0	65.50	10	31.0	37.75
os300_ie_2022_n300	10	**18**	27.5	25.00	12	17.5	25.25
os300_in_2022_n300	10	**19**	40.0 ▲	18.25	17	20.0 ▽	31.75
os300_it_2022_n300	10	**83**	53.0	38.50	52	38.0	47.25
os300_jp_2022_n300	10	32	45.5	46.75	**74**	43.5	38.25
os300_kr_2022_n300	10	**109**	33.0	54.25	10	21.5	39.00
os300_lt_2022_n300	10	18	51.0	67.25	**129**	44.0	39.25
os300_lu_2022_n300	10	89	29.5	28.75	**123**	25.5	61.00
os300_lv_2022_n300	10	**131**	66.0	31.75	**131**	38.5	64.50
os300_mt_2022_n300	10	**118**	57.0 ▲	43.00	27	21.5 ▽	28.00
os300_mx_2022_n300	10	30	40.0	30.25	**97**	28.5	34.75
os300_nl_2022_n300	10	32	30.5	37.50	**74**	33.0	47.75
os300_no_2022_n300	10	**52**	48.5 ▲	43.00	44	16.5 ▽	31.00
os300_pl_2022_n300	10	37	47.5	49.75	**44**	44.0	41.75
os300_pt_2022_n300	10	21	58.5 ▲	74.00	**59**	35.0 ▽	34.50
os300_ro_2022_n300	10	**51**	50.5	52.25	10	51.0	69.00
os300_ru_2022_n300	10	**41**	42.5	52.50	28	38.5	40.50
os300_se_2022_n300	10	**85**	63.0	52.00	32	51.5	33.75
os300_si_2022_n300	10	**101**	70.0	56.00	83	49.5	27.25
os300_sk_2022_n300	10	14	55.0	62.50	**58**	40.0	39.50
os300_tr_2022_n300	10	**98**	77.5	70.50	31	57.0	56.75
os300_tw_2022_n300	10	**25**	41.5	46.00	15	34.5	38.25
os300_us_2022_n300	10	**266**	33.0	66.00	10	39.5	51.00
os300_wa_2022_n300	10	**140**	47.5	38.50	16	43.5	37.50
os300_we_2022_n300	10	**60**	45.5	46.75	52	55.0	29.50
os300_wf_2022_n300	10	**43**	67.0 ▲	55.50	42	41.5 ▽	50.75
os300_wl_2022_n300	10	**52**	45.0	45.25	47	31.5	29.75
os300_wm_2022_n300	10	20	42.5	67.75	**224**	35.5	46.25
os300_za_2022_n300	10	**162**	38.0	64.25	14	44.5	50.75

Table 15: Results on MS-LOP for Δ_{NN} metric and $m = 15$ on os300 instances

Instance	**m**	MS-CD-RVNS			MS-MA-EDM		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
os300_at_2022_n300	15	**49**	81.0 ▲	71.75	19	35.5 ▽	46.50
os300_au_2022_n300	15	15	75.0	97.75	**79**	40.5	64.50
os300_be_2022_n300	15	**82**	58.0	49.00	18	63.0	54.75
os300_bg_2022_n300	15	78	93.0 ▲	58.50	**90**	64.5 ▽	58.75
os300_br_2022_n300	15	112	90.0 ▲	80.50	**146**	43.5 ▽	44.50
os300_ca_2022_n300	15	**83**	97.0 ▲	88.50	23	41.0 ▽	50.00
os300_ch_2022_n300	15	31	77.5	82.50	**70**	59.0	53.75
os300_cn_2022_n300	15	**124**	59.5	50.00	18	50.0	39.00
os300_cy_2022_n300	15	**211**	83.0 ▲	39.00	15	49.0 ▽	82.75
os300_cz_2022_n300	15	**87**	76.5	56.50	53	53.5	60.00
os300_de_2022_n300	15	29	70.5	73.00	**194**	65.0	66.50
os300_dk_2022_n300	15	39	60.0	69.25	**93**	52.5	57.75
os300_ee_2022_n300	15	16	78.5	51.50	**86**	79.0	44.00
os300_es_2022_n300	15	**75**	74.5 ▲	86.50	25	42.5 ▽	49.50
os300_fi_2022_n300	15	18	83.5	74.75	**77**	57.5	60.75
os300_fr_2022_n300	15	**104**	94.5	71.00	47	64.0	50.00
os300_gb_2022_n300	15	93	93.0	56.25	**107**	57.5	78.00
os300_gr_2022_n300	15	**195**	65.0	77.25	64	59.5	62.00
os300_hr_2022_n300	15	62	82.0 ▲	65.75	**75**	49.0 ▽	57.75
os300_hu_2022_n300	15	69	71.0	73.00	**84**	73.5	34.25
os300_id_2022_n300	15	**65**	87.5 ▲	98.50	22	53.0 ▽	58.50
os300_ie_2022_n300	15	**41**	41.0	33.50	25	25.0	35.75
os300_in_2022_n300	15	**38**	80.0 ▲	50.75	22	29.0 ▽	54.75
os300_it_2022_n300	15	151	89.5 ▲	58.50	**228**	53.5 ▽	66.25
os300_jp_2022_n300	15	**558**	72.5	78.25	101	66.5	61.50
os300_kr_2022_n300	15	**127**	57.0 ▲	76.75	19	33.0 ▽	50.00
os300_lt_2022_n300	15	116	89.0	59.50	**136**	77.0	53.25
os300_lu_2022_n300	15	105	44.5	42.50	**109**	27.0	57.25
os300_lv_2022_n300	15	**121**	99.5 ▲	65.25	110	60.0 ▽	49.00
os300_mt_2022_n300	15	**184**	82.5 ▲	44.75	30	31.5 ▽	39.25
os300_mx_2022_n300	15	**113**	68.5 ▲	57.25	**113**	43.5 ▽	52.75
os300_nl_2022_n300	15	42	59.5	58.00	**52**	48.0	43.25
os300_no_2022_n300	15	**93**	80.0 ▲	53.00	81	35.5 ▽	35.25
os300_pl_2022_n300	15	**115**	91.5 ▲	74.25	42	57.5 ▽	55.25
os300_pt_2022_n300	15	31	88.5 ▲	98.75	**78**	58.0 ▽	57.25
os300_ro_2022_n300	15	**72**	72.0	44.25	63	63.5	56.75
os300_ru_2022_n300	15	**52**	65.5	71.00	48	60.0	43.25
os300_se_2022_n300	15	**105**	87.5	64.75	64	79.0	48.00
os300_si_2022_n300	15	113	84.0	82.75	**123**	67.0	49.75
os300_sk_2022_n300	15	20	77.5	71.25	**91**	58.5	59.25
os300_tr_2022_n300	15	**103**	109.0	72.00	41	68.5	95.50
os300_tw_2022_n300	15	**39**	65.0	58.25	20	61.0	49.25
os300_us_2022_n300	15	**71**	67.5	71.25	18	53.5	56.50
os300_wa_2022_n300	15	**131**	77.5	76.00	17	68.0	49.50
os300_we_2022_n300	15	**100**	88.5	64.25	53	70.5	33.00
os300_wf_2022_n300	15	**86**	96.0 ▲	61.75	42	54.0 ▽	64.75
os300_wl_2022_n300	15	**66**	69.0 ▲	66.25	64	44.0 ▽	47.00
os300_wm_2022_n300	15	31	92.0 ▲	86.75	**120**	41.5 ▽	59.75
os300_za_2022_n300	15	**197**	62.5	80.00	15	53.0	67.50

Table 16: Results on MS-LOP for Δ_{NN} metric and $m = 5$ on PXP instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
pxp_at_2022_n155	5	206	207.0 ▽	32.00	**308**	300.0 ▲	33.50
pxp_au_2022_n155	5	232	291.0 ▽	85.00	**410**	427.5 ▲	64.75
pxp_be_2022_n155	5	413	395.5 ▽	68.50	**472**	539.5 ▲	74.75
pxp_bg_2022_n135	5	266	219.0 ▽	81.00	**333**	323.0 ▲	28.50
pxp_br_2022_n155	5	170	193.5	36.00	**208**	193.0	32.25
pxp_ca_2022_n158	5	559	571.5 ▽	27.00	**797**	757.0 ▲	61.75
pxp_ch_2022_n141	5	433	398.0 ▽	94.25	**745**	743.0 ▲	35.50
pxp_cn_2022_n169	5	496	482.5	75.75	**652**	479.0	133.75
pxp_cy_2022_n116	5	108	109.0 ▽	12.75	**170**	178.5 ▲	34.75
pxp_cz_2022_n162	5	473	383.5 ▽	98.75	**743**	722.0 ▲	72.00
pxp_de_2022_n165	5	255	234.5 ▽	65.50	**478**	485.0 ▲	92.75
pxp_dk_2022_n141	5	131	153.0 ▽	26.50	**200**	179.5 ▲	50.75
pxp_ee_2022_n126	5	257	239.5 ▽	84.00	**452**	450.5 ▲	30.25
pxp_es_2022_n162	5	193	252.5 ▽	77.50	**412**	401.5 ▲	53.75
pxp_fi_2022_n153	5	**199**	181.5 ▽	29.75	196	215.5 ▲	36.75
pxp_fr_2022_n164	5	636	531.0 ▽	163.25	**671**	671.5 ▲	148.50
pxp_gb_2022_n159	5	690	720.0 ▽	60.25	**703**	890.0 ▲	197.25
pxp_gr_2022_n130	5	772	771.0 ▽	146.00	**1185**	1172.5 ▲	102.75
pxp_hr_2022_n140	5	596	568.5 ▽	106.25	**956**	929.5 ▲	90.75
pxp_hu_2022_n160	5	207	189.5 ▽	61.25	**261**	294.0 ▲	60.00
pxp_id_2022_n132	5	**261**	262.0 ▲	28.75	181	239.0 ▽	55.00
pxp_ie_2022_n136	5	779	904.0 ▽	140.25	**1098**	1167.0 ▲	163.25
pxp_in_2022_n145	5	278	260.0 ▽	33.75	**325**	313.0 ▲	57.75
pxp_it_2022_n165	5	190	271.5 ▽	57.25	**291**	349.5 ▲	69.50
pxp_jp_2022_n159	5	156	119.5 ▽	30.50	**217**	188.5 ▲	26.50
pxp_kr_2022_n149	5	**251**	222.0	38.00	214	201.5	45.75
pxp_lt_2022_n133	5	275	366.0 ▽	65.25	**613**	540.0 ▲	62.75
pxp_lu_2022_n120	5	798	743.5 ▽	46.75	**838**	812.5 ▲	101.25
pxp_lv_2022_n132	5	291	258.5 ▽	58.75	**407**	415.5 ▲	37.75
pxp_mt_2022_n89	5	**404**	369.0 ▽	27.75	402	424.0 ▲	59.75
pxp_mx_2022_n139	5	158	157.5 ▽	28.50	**307**	286.0 ▲	32.75
pxp_nl_2022_n158	5	585	569.0 ▽	138.75	**659**	707.5 ▲	94.25
pxp_no_2022_n145	5	321	378.5 ▽	61.00	**512**	561.5 ▲	169.00
pxp_pl_2022_n166	5	182	223.0 ▽	57.50	**291**	319.0 ▲	31.25
pxp_pt_2022_n149	5	197	196.5 ▽	45.50	**217**	234.5 ▲	58.50
pxp_ro_2022_n151	5	215	344.0 ▽	38.50	**662**	466.0 ▲	83.00
pxp_ru_2022_n136	5	92	104.0 ▽	15.75	**161**	157.5 ▲	30.50
pxp_se_2022_n156	5	281	279.5 ▽	38.50	**380**	409.5 ▲	50.50
pxp_si_2022_n129	5	**237**	219.5 ▽	34.75	162	257.5 ▲	56.75
pxp_sk_2022_n151	5	**238**	219.5	35.50	190	213.5	48.00
pxp_tr_2022_n145	5	**188**	187.5 ▲	91.50	90	144.5 ▽	29.00
pxp_tw_2022_n138	5	**470**	391.5 ▽	45.00	425	478.0 ▲	88.25
pxp_us_2022_n170	5	320	326.5 ▽	57.50	**507**	495.0 ▲	42.50
pxp_wa_2022_n176	5	233	190.5 ▽	33.75	**326**	340.5 ▲	45.50
pxp_we_2022_n144	5	42	41.0 ▽	9.50	**70**	62.0 ▲	19.25
pxp_wf_2022_n141	5	111	90.0 ▽	16.00	**122**	112.5 ▲	17.50
pxp_wl_2022_n164	5	56	72.0 ▽	13.75	**133**	127.0 ▲	16.25
pxp_wm_2022_n161	5	355	274.0 ▽	59.75	**500**	518.0 ▲	33.75
pxp_za_2022_n134	5	**164**	154.0	18.50	160	170.5	43.25

Table 17: Results on MS-LOP for Δ_{NN} metric and $m = 10$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	10	317	313.5 ▽	51.00	**537**	504.0 ▲	70.75
pxp_au_2022_n155	10	349	457.5 ▽	159.00	**677**	728.0 ▲	114.75
pxp_be_2022_n155	10	630	605.5 ▽	127.75	**767**	909.0 ▲	146.50
pxp_bg_2022_n135	10	391	324.5 ▽	123.25	**560**	517.0 ▲	51.25
pxp_br_2022_n155	10	260	295.0 ▽	41.00	**340**	315.5 ▲	58.50
pxp_ca_2022_n158	10	907	955.5 ▽	81.75	**1348**	1338.5 ▲	99.75
pxp_ch_2022_n141	10	574	583.5 ▽	132.50	**1252**	1244.5 ▲	62.25
pxp_cn_2022_n169	10	685	712.5 ▽	82.25	**1047**	791.5 ▲	219.00
pxp_cy_2022_n116	10	161	163.0 ▽	19.50	**280**	291.5 ▲	66.75
pxp_cz_2022_n162	10	714	590.0 ▽	147.75	**1285**	1271.0 ▲	126.75
pxp_de_2022_n165	10	389	363.0 ▽	115.00	**811**	823.0 ▲	176.00
pxp_dk_2022_n141	10	200	224.0 ▽	56.75	**331**	292.0 ▲	79.50
pxp_ee_2022_n126	10	381	338.5 ▽	127.25	**756**	757.0 ▲	67.25
pxp_es_2022_n162	10	277	354.0 ▽	123.50	**715**	666.0 ▲	113.25
pxp_fi_2022_n153	10	274	265.5 ▽	48.50	**331**	354.0 ▲	47.50
pxp_fr_2022_n164	10	998	774.0 ▽	280.00	**1084**	1162.5 ▲	241.00
pxp_gb_2022_n159	10	1100	1095.0 ▽	188.00	**1249**	1541.0 ▲	293.25
pxp_gr_2022_n130	10	1079	1146.0 ▽	222.50	**2051**	2072.0 ▲	152.00
pxp_hr_2022_n140	10	877	895.0 ▽	175.50	**1696**	1642.0 ▲	177.00
pxp_hu_2022_n160	10	335	283.5 ▽	99.00	**419**	480.0 ▲	120.00
pxp_id_2022_n132	10	**406**	398.0	57.75	285	396.0	81.25
pxp_ie_2022_n136	10	1259	1419.0 ▽	221.75	**1806**	2049.0 ▲	311.25
pxp_in_2022_n145	10	423	400.5 ▽	55.75	**528**	517.5 ▲	90.25
pxp_it_2022_n165	10	260	389.0 ▽	108.00	**501**	569.0 ▲	129.75
pxp_jp_2022_n159	10	231	183.5 ▽	42.75	**362**	318.0 ▲	54.75
pxp_kr_2022_n149	10	**352**	316.5	41.50	339	303.0	68.50
pxp_lt_2022_n133	10	394	544.0 ▽	122.00	**1028**	895.0 ▲	96.75
pxp_lu_2022_n120	10	1187	1155.0 ▽	118.50	**1417**	1361.5 ▲	135.50
pxp_lv_2022_n132	10	450	379.0 ▽	94.00	**698**	701.5 ▲	58.75
pxp_mt_2022_n89	10	605	560.0 ▽	65.75	**656**	682.0 ▲	111.75
pxp_mx_2022_n139	10	243	233.5 ▽	42.00	**505**	471.0 ▲	61.25
pxp_nl_2022_n158	10	939	823.0 ▽	233.50	**1143**	1230.0 ▲	154.00
pxp_no_2022_n145	10	469	573.0 ▽	92.00	**888**	935.5 ▲	273.25
pxp_pl_2022_n166	10	253	360.5 ▽	69.75	**464**	532.5 ▲	53.50
pxp_pt_2022_n149	10	280	282.0 ▽	75.00	**321**	375.5 ▲	90.25
pxp_ro_2022_n151	10	308	545.5 ▽	75.50	**1099**	773.5 ▲	150.25
pxp_ru_2022_n136	10	149	150.5 ▽	32.50	**250**	251.5 ▲	66.25
pxp_se_2022_n156	10	422	416.5 ▽	76.00	**654**	662.5 ▲	81.50
pxp_si_2022_n129	10	**363**	337.5 ▽	73.00	260	409.0 ▲	91.25
pxp_sk_2022_n151	10	**372**	318.0	68.50	293	330.0	78.50
pxp_tr_2022_n145	10	**220**	253.0	89.25	134	224.0	56.00
pxp_tw_2022_n138	10	**780**	606.5 ▽	85.25	728	828.5 ▲	156.75
pxp_us_2022_n170	10	521	544.5 ▽	81.25	**873**	868.5 ▲	73.75
pxp_wa_2022_n176	10	390	291.0 ▽	79.25	**564**	568.5 ▲	86.50
pxp_we_2022_n144	10	54	54.0 ▽	18.25	**102**	91.5 ▲	33.50
pxp_wf_2022_n141	10	182	138.0 ▽	26.50	**189**	178.5 ▲	25.00
pxp_wl_2022_n164	10	88	102.0 ▽	15.75	**219**	206.5 ▲	31.00
pxp_wm_2022_n161	10	557	423.0 ▽	95.00	**862**	889.0 ▲	51.75
pxp_za_2022_n134	10	264	241.5 ▽	33.00	**274**	283.5 ▲	72.50

Table 18: Results on MS-LOP for Δ_{NN} metric and $m = 15$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	15	438	404.0 ▽	98.00	**730**	692.0 ▲	87.00
pxp_au_2022_n155	15	473	576.0 ▽	223.75	**922**	1010.0 ▲	169.50
pxp_be_2022_n155	15	797	756.5 ▽	161.25	**1060**	1259.0 ▲	209.25
pxp_bg_2022_n135	15	484	399.0 ▽	185.50	**750**	705.0 ▲	72.75
pxp_br_2022_n155	15	333	378.0 ▽	54.50	**462**	432.0 ▲	87.00
pxp_ca_2022_n158	15	1186	1252.5 ▽	166.25	**1916**	1874.0 ▲	151.75
pxp_ch_2022_n141	15	678	718.0 ▽	191.25	**1695**	1703.0 ▲	77.75
pxp_cn_2022_n169	15	864	881.5 ▽	138.50	**1424**	1070.5 ▲	291.75
pxp_cy_2022_n116	15	212	206.0 ▽	23.00	**376**	395.5 ▲	100.75
pxp_cz_2022_n162	15	943	733.5 ▽	223.00	**1796**	1756.0 ▲	180.50
pxp_de_2022_n165	15	523	457.5 ▽	161.00	**1112**	1153.5 ▲	239.00
pxp_dk_2022_n141	15	240	281.5 ▽	81.75	**452**	395.0 ▲	113.25
pxp_ee_2022_n126	15	481	428.5 ▽	206.00	**1025**	1053.5 ▲	74.00
pxp_es_2022_n162	15	333	470.5 ▽	122.75	**986**	903.5 ▲	166.25
pxp_fi_2022_n153	15	365	347.5 ▽	64.00	**462**	488.0 ▲	59.25
pxp_fr_2022_n164	15	1350	1037.5 ▽	380.50	**1530**	1632.0 ▲	356.25
pxp_gb_2022_n159	15	1433	1390.0 ▽	194.00	**1758**	2107.0 ▲	397.75
pxp_gr_2022_n130	15	1396	1439.5 ▽	293.25	**2904**	2919.5 ▲	276.25
pxp_hr_2022_n140	15	1071	1178.5 ▽	285.00	**2378**	2311.5 ▲	233.00
pxp_hu_2022_n160	15	425	322.5 ▽	125.00	**566**	652.5 ▲	165.00
pxp_id_2022_n132	15	**511**	506.0 ▽	57.75	392	536.0 ▲	109.25
pxp_ie_2022_n136	15	1661	1874.0 ▽	282.50	**2477**	2873.0 ▲	409.00
pxp_in_2022_n145	15	569	502.0 ▽	64.50	**700**	694.0 ▲	135.25
pxp_it_2022_n165	15	323	477.0 ▽	145.50	**700**	779.0 ▲	164.25
pxp_jp_2022_n159	15	311	235.5 ▽	60.25	**488**	433.0 ▲	91.75
pxp_kr_2022_n149	15	444	387.5	78.75	**449**	398.5	102.00
pxp_lt_2022_n133	15	469	701.5 ▽	158.50	**1448**	1235.0 ▲	143.25
pxp_lu_2022_n120	15	1548	1540.0 ▽	183.00	**1934**	1895.5 ▲	201.00
pxp_lv_2022_n132	15	573	454.0 ▽	112.75	**936**	975.0 ▲	79.25
pxp_mt_2022_n89	15	758	746.5 ▽	77.25	**865**	926.5 ▲	152.25
pxp_mx_2022_n139	15	307	286.0 ▽	44.75	**685**	645.5 ▲	92.50
pxp_nl_2022_n158	15	1160	1107.5 ▽	324.50	**1613**	1726.5 ▲	217.25
pxp_no_2022_n145	15	581	746.5 ▽	130.00	**1214**	1295.5 ▲	370.50
pxp_pl_2022_n166	15	337	459.5 ▽	123.50	**628**	722.0 ▲	86.00
pxp_pt_2022_n149	15	368	348.5 ▽	86.25	**411**	499.5 ▲	134.00
pxp_ro_2022_n151	15	403	696.5 ▽	95.25	**1490**	1067.0 ▲	224.75
pxp_ru_2022_n136	15	190	183.5 ▽	50.75	**325**	332.0 ▲	90.00
pxp_se_2022_n156	15	558	549.5 ▽	108.25	**887**	904.0 ▲	105.75
pxp_si_2022_n129	15	**496**	421.0 ▽	86.75	342	548.5 ▲	115.00
pxp_sk_2022_n151	15	**490**	418.5	78.50	397	426.5	118.75
pxp_tr_2022_n145	15	**264**	275.0	114.25	169	285.0	71.00
pxp_tw_2022_n138	15	**1066**	783.0 ▽	119.50	988	1149.0 ▲	215.00
pxp_us_2022_n170	15	697	713.0 ▽	131.25	**1230**	1212.0 ▲	102.25
pxp_wa_2022_n176	15	524	374.5 ▽	113.00	**764**	790.5 ▲	129.25
pxp_we_2022_n144	15	66	61.5 ▽	25.25	**132**	121.5 ▲	46.50
pxp_wf_2022_n141	15	236	176.0 ▽	36.25	**251**	239.0 ▲	40.00
pxp_wl_2022_n164	15	114	124.5 ▽	19.25	**295**	280.5 ▲	41.25
pxp_wm_2022_n161	15	658	534.5 ▽	142.50	**1195**	1237.5 ▲	76.00
pxp_za_2022_n134	15	357	313.5 ▽	52.50	**376**	386.5 ▲	109.25

Table 19: Results on MS-LOP for Δ_{NN} metric and $m = 5$ on RXR instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
rxr__1995__n49	5	**19**	19.0	0.00	**19**	19.0	0.00
rxr__1996__n49	5	50	50.0 ▽	0.00	**51**	51.0 ▲	1.00
rxr__1997__n49	5	**31**	31.0 ▲	34.00	23	23.0 ▽	0.00
rxr__1998__n49	5	**47**	47.0	0.00	**47**	47.0	0.00
rxr__1999__n49	5	**19**	19.0	0.00	**19**	19.0	0.00
rxr__2000__n49	5	**15**	15.0	0.00	**15**	15.0	0.00
rxr__2001__n49	5	**8**	8.0	0.00	**8**	8.0	0.00
rxr__2002__n49	5	**39**	39.0	0.00	**39**	39.0	0.00
rxr__2003__n49	5	**38**	38.0	0.00	**38**	38.0	0.00
rxr__2004__n49	5	**31**	31.0	0.00	**31**	31.0	0.00
rxr__2005__n49	5	**58**	40.0	18.00	40	40.0	0.00
rxr__2006__n49	5	**36**	36.0 ▲	20.00	35	35.0 ▽	0.00
rxr__2007__n49	5	**39**	39.0	0.00	**39**	39.0	0.00
rxr__2008__n49	5	**33**	33.0	0.00	**33**	33.0	0.00
rxr__2009__n49	5	**8**	8.0	0.00	**8**	8.0	0.00
rxr__2010__n49	5	**38**	38.0	0.00	**38**	38.0	0.00
rxr__2011__n49	5	**29**	29.0	1.00	**29**	29.0	0.00
rxr__2012__n49	5	**31**	31.0	0.00	**31**	31.0	0.00
rxr__2013__n49	5	**54**	54.0	0.00	**54**	54.0	0.00
rxr__2014__n49	5	**43**	43.0 ▲	1.75	41	41.0 ▽	0.00
rxr__2015__n49	5	**20**	20.0	0.00	**20**	20.0	0.00
rxr__2016__n49	5	**26**	26.0 ▲	5.00	21	21.0 ▽	0.00
rxr__2017__n49	5	77	83.0	7.00	**78**	76.5	19.50
rxr__2018__n49	5	**35**	35.0	0.00	**35**	35.0	0.00
rxr__2019__n49	5	**63**	63.0 ▲	11.25	28	28.0 ▽	0.00
rxr__2020__n49	5	**28**	28.0	0.00	**28**	28.0	0.00
rxr__2021__n49	5	**53**	46.5 ▽	13.00	**53**	53.0 ▲	0.00
rxr__2022__n49	5	**24**	24.0	0.00	**24**	24.0	0.00

Table 20: Results on MS-LOP for Δ_{NN} metric and $m = 10$ on RXR instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
rxr_1995_n49	10	**50**	59.0 ▲	15.00	42	42.0 ▽	0.00
rxr_1996_n49	10	50	50.0 ▽	11.00	**68**	68.0 ▲	7.00
rxr_1997_n49	10	**61**	42.0 ▲	53.00	10	10.0 ▽	14.00
rxr_1998_n49	10	**90**	83.0 ▲	12.00	47	56.0 ▽	9.00
rxr_1999_n49	10	**41**	41.0 ▲	14.00	30	30.0 ▽	0.00
rxr_2000_n49	10	**20**	20.0	0.00	**20**	20.0	3.00
rxr_2001_n49	10	**35**	35.0	0.00	31	35.0	9.25
rxr_2002_n49	10	56	56.0	0.00	**59**	56.0	0.75
rxr_2003_n49	10	**84**	80.0 ▲	20.00	51	38.0 ▽	25.25
rxr_2004_n49	10	**61**	61.0	0.00	**61**	61.0	0.00
rxr_2005_n49	10	**91**	90.0 ▲	1.00	90	84.0 ▽	21.50
rxr_2006_n49	10	46	59.5 ▲	11.00	**57**	48.0 ▽	9.75
rxr_2007_n49	10	**55**	38.0	17.00	38	38.0	1.00
rxr_2008_n49	10	70	70.0 ▲	5.00	**75**	65.0 ▽	3.50
rxr_2009_n49	10	**36**	21.0 ▲	26.00	10	10.0 ▽	0.00
rxr_2010_n49	10	**90**	83.5 ▲	8.00	54	70.0 ▽	16.00
rxr_2011_n49	10	29	29.0 ▽	1.00	**30**	30.0 ▲	0.00
rxr_2012_n49	10	**36**	36.0 ▲	2.00	31	31.0 ▽	8.00
rxr_2013_n49	10	**78**	78.0	0.00	74	78.0	6.75
rxr_2014_n49	10	57	59.0 ▽	8.00	**72**	72.0 ▲	8.75
rxr_2015_n49	10	32	54.0	9.50	**77**	46.0	19.50
rxr_2016_n49	10	**43**	42.5 ▲	5.75	20	23.0 ▽	8.50
rxr_2017_n49	10	**103**	115.5 ▲	18.50	78	79.0 ▽	20.00
rxr_2018_n49	10	**54**	56.0	14.00	53	56.0	15.25
rxr_2019_n49	10	**63**	63.0 ▲	11.25	28	28.0 ▽	0.00
rxr_2020_n49	10	**64**	60.0 ▲	10.00	52	52.0 ▽	1.00
rxr_2021_n49	10	53	46.5 ▽	13.00	**60**	53.0 ▲	7.00
rxr_2022_n49	10	40	40.0	0.00	**52**	40.0	7.50

Table 21: Results on MS-LOP for Δ_{NN} metric and $m = 15$ on RXR instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
rxr_1995_n49	15	**131**	112.0 ▲	30.00	59	69.0 ▽	18.00
rxr_1996_n49	15	50	50.0 ▽	11.00	**68**	68.0 ▲	7.00
rxr_1997_n49	15	**114**	71.0 ▲	52.75	26	28.0 ▽	10.00
rxr_1998_n49	15	**90**	83.0 ▲	12.00	47	56.0 ▽	9.00
rxr_1999_n49	15	51	56.0	6.75	**55**	56.0	9.00
rxr_2000_n49	15	**37**	32.0	4.00	33	30.0	6.25
rxr_2001_n49	15	**43**	43.0 ▲	0.00	31	39.0 ▽	5.50
rxr_2002_n49	15	84	82.0 ▽	2.00	**98**	86.0 ▲	7.00
rxr_2003_n49	15	**84**	80.0 ▲	20.00	51	38.0 ▽	25.25
rxr_2004_n49	15	106	96.5	10.75	**114**	93.0	12.25
rxr_2005_n49	15	**109**	109.0 ▲	5.00	106	99.0 ▽	19.25
rxr_2006_n49	15	46	59.5 ▲	11.00	**57**	48.0 ▽	9.75
rxr_2007_n49	15	**55**	38.0	17.00	38	38.0	1.00
rxr_2008_n49	15	102	102.0	12.25	**105**	103.5	6.75
rxr_2009_n49	15	**49**	38.0 ▲	42.50	25	25.0 ▽	0.00
rxr_2010_n49	15	**90**	84.0 ▲	11.75	54	70.0 ▽	16.00
rxr_2011_n49	15	29	29.0 ▽	1.00	**30**	30.0 ▲	0.00
rxr_2012_n49	15	**36**	36.0 ▲	2.00	31	31.0 ▽	8.00
rxr_2013_n49	15	**109**	109.0 ▲	17.00	74	79.0 ▽	11.00
rxr_2014_n49	15	57	59.0 ▽	8.00	**85**	81.0 ▲	11.00
rxr_2015_n49	15	**79**	54.0	17.00	77	46.0	19.50
rxr_2016_n49	15	**43**	42.5 ▲	5.75	31	31.0 ▽	9.00
rxr_2017_n49	15	**103**	116.5 ▲	20.50	78	79.0 ▽	20.00
rxr_2018_n49	15	**101**	86.0 ▲	15.50	88	77.5 ▽	15.50
rxr_2019_n49	15	**63**	63.0 ▲	11.25	28	28.0 ▽	0.00
rxr_2020_n49	15	71	69.0 ▽	7.00	**88**	84.5 ▲	12.00
rxr_2021_n49	15	53	46.5 ▽	13.00	**60**	53.0 ▲	7.00
rxr_2022_n49	15	**70**	70.0 ▲	5.00	52	53.0 ▽	7.50

Table 22: Results on MS-LOP for Δ_{SP} metric and $m = 5$ on OS300 instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
os300_at_2022_n300	5	**4.4**	4.52	0.58	4.26	4.64	0.59
os300_au_2022_n300	5	4.49	4.57	0.59	**5.0**	4.4	0.61
os300_be_2022_n300	5	**4.68**	4.68	0.59	3.89	4.97	0.60
os300_bg_2022_n300	5	**5.0**	4.99	0.31	**5.0**	4.98	0.60
os300_br_2022_n300	5	4.63	4.69	0.59	**5.0**	4.4	0.58
os300_ca_2022_n300	5	**5.0**	4.54	0.66	3.89	4.52	0.61
os300_ch_2022_n300	5	3.89	4.7	0.59	**5.0**	4.98	0.60
os300_cn_2022_n300	5	**4.4**	4.4	0.61	4.38	4.55	0.60
os300_cy_2022_n300	5	**5.0**	4.98	0.52	3.94	4.88	0.60
os300_cz_2022_n300	5	**5.0**	4.84	0.61	4.99	4.99	0.60
os300_de_2022_n300	5	4.38	4.72	0.60	**5.0**	4.98	0.60
os300_dk_2022_n300	5	4.4	4.4	0.64	**5.0**	4.54	0.60
os300_ee_2022_n300	5	4.34	4.69	0.60	**4.98**	5.0	0.03
os300_es_2022_n300	5	**4.4**	4.95	0.59	**4.4**	4.62	0.59
os300_fi_2022_n300	5	3.94	4.7	0.65	**5.0**	4.96	0.69
os300_fr_2022_n300	5	**4.4**	4.82	0.60	**4.4**	4.79	0.60
os300_gb_2022_n300	5	4.69	4.7	0.59	**5.0**	4.98	0.60
os300_gr_2022_n300	5	**5.0**	4.4	0.57	**5.0**	4.69	0.61
os300_hr_2022_n300	5	**5.0**	4.7	0.60	**5.0**	4.79	0.60
os300_hu_2022_n300	5	**4.97**	4.7	0.59	4.4	4.97	0.54
os300_id_2022_n300	5	**4.7**	4.7	0.58	3.89	4.64	0.60
os300_ie_2022_n300	5	**4.34**	4.36	0.81	3.94	4.2	0.87
os300_in_2022_n300	5	**4.08**	4.4	0.31	3.89	4.4	0.93
os300_it_2022_n300	5	**5.0**	4.93	0.60	**5.0**	4.64	0.60
os300_jp_2022_n300	5	4.97	4.7	0.61	**5.0**	4.78	0.60
os300_kr_2022_n300	5	**4.4**	4.54	0.48	4.38	4.4	0.61
os300_lt_2022_n300	5	4.69	4.69	0.60	**5.0**	4.99	0.60
os300_lu_2022_n300	5	4.4	4.38	0.60	**5.0**	4.4	0.61
os300_lv_2022_n300	5	**5.0**	4.98	0.60	**5.0**	4.98	0.60
os300_mt_2022_n300	5	**5.0**	4.69	0.43	3.89	4.4	0.97
os300_mx_2022_n300	5	4.95	4.63	0.57	**5.0**	4.79	0.60
os300_nl_2022_n300	5	4.4	4.4	0.55	**4.98**	4.98	0.60
os300_no_2022_n300	5	4.87	4.7	0.58	**5.0**	4.38	1.01
os300_pl_2022_n300	5	**4.97**	4.87	0.31	**4.97**	4.97	0.60
os300_pt_2022_n300	5	3.59	4.7	0.61	**5.0**	4.96	0.60
os300_ro_2022_n300	5	**4.99**	4.68	0.72	4.38	4.84	0.60
os300_ru_2022_n300	5	**4.98**	4.69	0.58	4.4	4.84	0.60
os300_se_2022_n300	5	**5.0**	4.84	0.60	4.98	4.99	0.46
os300_si_2022_n300	5	4.98	4.95	0.60	**5.0**	4.98	0.46
os300_sk_2022_n300	5	4.45	4.57	0.59	**5.0**	4.88	0.60
os300_tr_2022_n300	5	4.13	4.79	0.31	**4.69**	4.69	0.60
os300_tw_2022_n300	5	**4.4**	4.54	0.67	4.1	4.91	0.60
os300_us_2022_n300	5	**4.7**	4.69	0.46	4.39	4.55	0.68
os300_wa_2022_n300	5	**5.0**	4.7	0.60	4.38	4.84	0.60
os300_we_2022_n300	5	**4.99**	4.96	0.55	**4.99**	4.99	0.02
os300_wf_2022_n300	5	**5.0**	4.99	0.60	4.99	4.99	0.60
os300_wl_2022_n300	5	4.4	4.69	0.60	**5.0**	4.4	0.61
os300_wm_2022_n300	5	**4.38**	4.7	0.65	3.89	4.4	0.63
os300_za_2022_n300	5	**4.4**	4.4	0.30	4.08	4.78	0.60

Table 23: Results on MS-LOP for Δ_{SP} metric and $m = 10$ on OS300 instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
os300_at_2022_n300	10	**9.09**	9.1	1.16	7.59	9.39	1.45
os300_au_2022_n300	10	6.25	8.8	1.73	**10.0**	8.79	1.51
os300_be_2022_n300	10	**9.67**	9.37	1.40	7.71	9.72	1.12
os300_bg_2022_n300	10	9.99	9.98	0.53	**10.0**	9.94	1.20
os300_br_2022_n300	10	9.6	9.5	0.89	**10.0**	8.8	1.47
os300_ca_2022_n300	10	**10.0**	9.23	1.41	7.71	8.93	1.50
os300_ch_2022_n300	10	7.99	9.4	0.91	**9.99**	9.95	1.20
os300_cn_2022_n300	10	**9.09**	9.08	1.15	8.76	9.4	1.46
os300_cy_2022_n300	10	**9.99**	9.96	0.60	6.9	9.72	1.28
os300_cz_2022_n300	10	**9.97**	9.54	1.13	9.96	9.81	1.23
os300_de_2022_n300	10	8.75	9.62	1.18	**10.0**	9.95	1.20
os300_dk_2022_n300	10	8.8	9.09	1.45	**10.0**	9.38	1.25
os300_ee_2022_n300	10	8.61	9.38	1.18	**9.95**	9.98	0.05
os300_es_2022_n300	10	**8.8**	9.77	0.88	8.5	9.21	1.20
os300_fi_2022_n300	10	7.39	9.4	1.42	**10.0**	9.93	1.30
os300_fr_2022_n300	10	**9.1**	9.77	0.92	8.8	9.73	1.19
os300_gb_2022_n300	10	9.38	9.4	0.82	**10.0**	9.92	1.42
os300_gr_2022_n300	10	**9.98**	8.88	1.17	9.97	9.38	1.40
os300_hr_2022_n300	10	9.99	9.33	0.99	**10.0**	9.55	1.43
os300_hu_2022_n300	10	**9.91**	9.66	0.87	9.39	9.91	1.12
os300_id_2022_n300	10	**9.7**	9.69	0.90	7.69	9.58	1.24
os300_ie_2022_n300	10	**8.5**	8.67	0.85	6.85	8.29	1.27
os300_in_2022_n300	10	**8.46**	9.19	0.75	7.99	8.8	1.75
os300_it_2022_n300	10	**10.0**	9.82	0.89	9.99	9.4	1.48
os300_jp_2022_n300	10	9.95	9.39	1.19	**10.0**	9.67	1.25
os300_kr_2022_n300	10	**9.06**	9.1	1.37	8.46	8.78	1.50
os300_lt_2022_n300	10	9.07	9.38	1.21	**10.0**	9.95	1.22
os300_lu_2022_n300	10	9.4	8.73	0.87	**10.0**	8.79	1.53
os300_lv_2022_n300	10	9.99	9.98	0.92	**10.0**	9.94	1.22
os300_mt_2022_n300	10	**9.99**	9.38	1.21	7.77	8.79	1.91
os300_mx_2022_n300	10	9.91	9.22	1.36	**10.0**	9.28	1.19
os300_nl_2022_n300	10	8.8	8.8	1.10	**9.96**	9.93	1.23
os300_no_2022_n300	10	9.79	9.39	1.05	**9.99**	8.64	1.94
os300_pl_2022_n300	10	**9.95**	9.74	0.83	9.64	9.84	1.24
os300_pt_2022_n300	10	7.99	9.68	1.24	**10.0**	9.89	1.22
os300_ro_2022_n300	10	**9.94**	9.1	1.05	8.46	9.8	1.50
os300_ru_2022_n300	10	**9.96**	9.24	1.36	8.79	9.66	1.42
os300_se_2022_n300	10	**10.0**	9.67	0.92	9.94	9.98	0.82
os300_si_2022_n300	10	9.96	9.85	0.90	**9.99**	9.97	0.70
os300_sk_2022_n300	10	8.11	9.1	1.16	**10.0**	9.76	1.20
os300_tr_2022_n300	10	8.11	9.65	0.64	**9.38**	9.24	1.21
os300_tw_2022_n300	10	**9.09**	9.38	1.04	7.99	9.78	1.24
os300_us_2022_n300	10	**9.39**	9.36	0.93	8.48	9.1	1.90
os300_wa_2022_n300	10	**9.99**	9.38	1.09	8.76	9.54	1.23
os300_we_2022_n300	10	**9.98**	9.91	0.68	9.97	9.98	0.04
os300_wf_2022_n300	10	**9.99**	9.89	0.65	9.98	9.96	1.24
os300_wl_2022_n300	10	9.09	9.39	1.35	**9.98**	8.8	1.23
os300_wm_2022_n300	10	**9.06**	9.37	1.33	7.78	8.88	1.39
os300_za_2022_n300	10	**8.8**	9.08	0.81	8.48	9.31	1.19

Table 24: Results on MS-LOP for Δ_{SP} metric and $m = 15$ on OS300 instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
os300_at_2022_n300	15	**14.09**	14.09	1.36	11.82	14.01	2.05
os300_au_2022_n300	15	9.83	13.65	2.12	**14.99**	13.32	2.12
os300_be_2022_n300	15	**14.36**	13.92	1.95	11.54	14.7	1.80
os300_bg_2022_n300	15	14.69	14.95	0.77	**14.98**	14.76	1.73
os300_br_2022_n300	15	14.28	14.39	0.95	**15.0**	13.49	2.09
os300_ca_2022_n300	15	**14.99**	14.22	1.99	11.57	13.61	2.08
os300_ch_2022_n300	15	12.39	14.37	1.48	**14.97**	14.81	1.82
os300_cn_2022_n300	15	**13.78**	13.78	2.42	12.84	13.94	2.08
os300_cy_2022_n300	15	**14.99**	14.93	0.63	9.48	14.66	1.98
os300_cz_2022_n300	15	**14.95**	14.37	1.66	14.93	14.78	2.11
os300_de_2022_n300	15	13.14	14.41	1.48	**15.0**	14.92	1.80
os300_dk_2022_n300	15	13.49	13.77	2.04	**14.99**	13.92	1.81
os300_ee_2022_n300	15	11.92	14.24	1.50	**14.92**	14.96	0.08
os300_es_2022_n300	15	**13.49**	14.69	1.17	12.9	13.92	1.82
os300_fi_2022_n300	15	10.98	14.38	2.03	**14.99**	14.75	2.14
os300_fr_2022_n300	15	**14.09**	14.68	1.05	13.2	14.34	1.71
os300_gb_2022_n300	15	14.07	14.39	1.08	**14.99**	14.87	2.09
os300_gr_2022_n300	15	**14.98**	13.75	1.77	14.95	14.05	2.05
os300_hr_2022_n300	15	14.98	14.23	1.22	**15.0**	14.3	2.09
os300_hu_2022_n300	15	**14.84**	14.48	1.12	14.38	14.81	1.72
os300_id_2022_n300	15	**14.69**	14.52	1.20	11.58	14.51	2.01
os300_ie_2022_n300	15	**13.19**	13.04	1.82	10.5	12.42	2.42
os300_in_2022_n300	15	**12.86**	13.85	1.28	12.09	13.19	2.30
os300_it_2022_n300	15	**15.0**	14.81	1.19	14.98	14.07	2.10
os300_jp_2022_n300	15	14.94	13.8	1.79	**15.0**	14.49	1.80
os300_kr_2022_n300	15	**14.01**	14.08	2.18	12.86	13.2	2.09
os300_lt_2022_n300	15	14.05	14.19	1.51	**15.0**	14.91	1.80
os300_lu_2022_n300	15	14.39	13.18	1.49	**15.0**	13.16	2.87
os300_lv_2022_n300	15	14.97	14.96	0.97	**15.0**	14.92	1.83
os300_mt_2022_n300	15	**14.99**	14.07	1.52	11.52	13.19	2.99
os300_mx_2022_n300	15	14.88	13.79	1.88	**15.0**	13.8	2.05
os300_nl_2022_n300	15	13.49	13.74	1.72	**14.93**	14.88	1.85
os300_no_2022_n300	15	14.73	14.09	1.52	**14.99**	13.14	2.82
os300_pl_2022_n300	15	**14.93**	14.67	1.12	14.01	14.36	1.81
os300_pt_2022_n300	15	12.05	14.37	1.88	**14.98**	14.82	1.82
os300_ro_2022_n300	15	**14.93**	13.8	1.51	13.46	14.63	2.02
os300_ru_2022_n300	15	**14.93**	13.79	1.49	13.19	14.62	1.79
os300_se_2022_n300	15	**15.0**	14.53	1.20	14.92	14.96	1.12
os300_si_2022_n300	15	14.93	14.83	1.12	**14.99**	14.94	0.70
os300_sk_2022_n300	15	12.34	13.75	1.46	**14.99**	14.61	1.79
os300_tr_2022_n300	15	11.78	14.5	1.11	**14.04**	13.94	2.04
os300_tw_2022_n300	15	**13.78**	14.08	1.45	12.09	14.66	1.70
os300_us_2022_n300	15	**14.38**	14.09	1.11	12.88	13.94	2.84
os300_wa_2022_n300	15	**14.98**	14.08	1.63	12.82	14.24	1.91
os300_we_2022_n300	15	**14.98**	14.89	1.20	14.94	14.96	0.06
os300_wf_2022_n300	15	**14.99**	14.72	1.00	14.07	14.76	1.80
os300_wl_2022_n300	15	13.77	14.06	1.81	**14.96**	13.46	2.00
os300_wm_2022_n300	15	**13.74**	14.05	1.61	11.57	13.46	2.10
os300_za_2022_n300	15	**13.49**	13.78	1.20	12.58	13.92	1.98

Table 25: Results on MS-LOP for Δ_{SP} metric and $m = 5$ on PXP instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
pxp_at_2022_n155	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_au_2022_n155	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_be_2022_n155	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_bg_2022_n135	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_br_2022_n155	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_ca_2022_n158	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_ch_2022_n141	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_cn_2022_n169	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_cy_2022_n116	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_cz_2022_n162	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_de_2022_n165	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_dk_2022_n141	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_ee_2022_n126	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_es_2022_n162	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_fi_2022_n153	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_fr_2022_n164	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_gb_2022_n159	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_gr_2022_n130	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_hr_2022_n140	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_hu_2022_n160	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_id_2022_n132	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_ie_2022_n136	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_in_2022_n145	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_it_2022_n165	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_jp_2022_n159	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_kr_2022_n149	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_lt_2022_n133	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_lu_2022_n120	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_lv_2022_n132	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_mt_2022_n89	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_mx_2022_n139	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_nl_2022_n158	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_no_2022_n145	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_pl_2022_n166	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_pt_2022_n149	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_ro_2022_n151	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_ru_2022_n136	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_se_2022_n156	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_si_2022_n129	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_sk_2022_n151	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_tr_2022_n145	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_tw_2022_n138	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_us_2022_n170	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_wa_2022_n176	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_we_2022_n144	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_wf_2022_n141	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_wl_2022_n164	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_wm_2022_n161	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
pxp_za_2022_n134	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00

Table 26: Results on MS-LOP for Δ_{SP} metric and $m = 10$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_au_2022_n155	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_be_2022_n155	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_bg_2022_n135	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_br_2022_n155	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_ca_2022_n158	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_ch_2022_n141	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_cn_2022_n169	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_cy_2022_n116	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_cz_2022_n162	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_de_2022_n165	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_dk_2022_n141	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_ee_2022_n126	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_es_2022_n162	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_fi_2022_n153	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_fr_2022_n164	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_gb_2022_n159	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_gr_2022_n130	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_hr_2022_n140	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_hu_2022_n160	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_id_2022_n132	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_ie_2022_n136	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_in_2022_n145	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_it_2022_n165	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_jp_2022_n159	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_kr_2022_n149	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_lt_2022_n133	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_lu_2022_n120	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_lv_2022_n132	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_mt_2022_n89	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_mx_2022_n139	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_nl_2022_n158	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_no_2022_n145	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_pl_2022_n166	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_pt_2022_n149	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_ro_2022_n151	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_ru_2022_n136	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_se_2022_n156	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_si_2022_n129	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_sk_2022_n151	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_tr_2022_n145	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_tw_2022_n138	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_us_2022_n170	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_wa_2022_n176	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_we_2022_n144	10	**10.0**	10.0	0.01	**10.0**	10.0	0.00
pxp_wf_2022_n141	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_wl_2022_n164	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_wm_2022_n161	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
pxp_za_2022_n134	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00

Table 27: Results on MS-LOP for Δ_{SP} metric and $m = 15$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_au_2022_n155	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_be_2022_n155	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_bg_2022_n135	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_br_2022_n155	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_ca_2022_n158	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_ch_2022_n141	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_cn_2022_n169	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_cy_2022_n116	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_cz_2022_n162	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_de_2022_n165	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_dk_2022_n141	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_ee_2022_n126	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_es_2022_n162	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_fi_2022_n153	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_fr_2022_n164	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_gb_2022_n159	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_gr_2022_n130	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_hr_2022_n140	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_hu_2022_n160	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_id_2022_n132	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_ie_2022_n136	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_in_2022_n145	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_it_2022_n165	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_jp_2022_n159	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_kr_2022_n149	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_lt_2022_n133	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_lu_2022_n120	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_lv_2022_n132	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_mt_2022_n89	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_mx_2022_n139	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_nl_2022_n158	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_no_2022_n145	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_pl_2022_n166	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_pt_2022_n149	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_ro_2022_n151	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_ru_2022_n136	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_se_2022_n156	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_si_2022_n129	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_sk_2022_n151	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_tr_2022_n145	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_tw_2022_n138	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_us_2022_n170	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_wa_2022_n176	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_we_2022_n144	15	14.98	14.96 ▽	0.12	**15.0**	15.0 ▲	0.00
pxp_wf_2022_n141	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_wl_2022_n164	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_wm_2022_n161	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
pxp_za_2022_n134	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00

Table 28: Results on MS-LOP for Δ_{SP} metric and $m = 5$ on RXR instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
rxr_1995_n49	5	**4.97**	4.97	0.00	**4.97**	4.97	0.00
rxr_1996_n49	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
rxr_1997_n49	5	**4.4**	4.4	0.30	**4.4**	4.4	0.00
rxr_1998_n49	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
rxr_1999_n49	5	**4.96**	4.96	0.00	**4.96**	4.96	0.00
rxr_2000_n49	5	**4.69**	4.69	0.00	**4.69**	4.69	0.00
rxr_2001_n49	5	**4.4**	4.4	0.00	**4.4**	4.4	0.00
rxr_2002_n49	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
rxr_2003_n49	5	**4.98**	4.98	0.00	**4.98**	4.98	0.00
rxr_2004_n49	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
rxr_2005_n49	5	**5.0**	4.99	0.01	4.99	4.99	0.00
rxr_2006_n49	5	**4.7**	4.7	0.30	**4.7**	4.7	0.00
rxr_2007_n49	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
rxr_2008_n49	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
rxr_2009_n49	5	**4.4**	4.4	0.00	**4.4**	4.4	0.00
rxr_2010_n49	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
rxr_2011_n49	5	**5.0**	5.0	1.00	**5.0**	5.0	0.00
rxr_2012_n49	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
rxr_2013_n49	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
rxr_2014_n49	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
rxr_2015_n49	5	**4.99**	4.99	0.00	**4.99**	4.99	0.00
rxr_2016_n49	5	**4.7**	4.7 ▲	0.00	4.4	4.4 ▽	0.00
rxr_2017_n49	5	**5.0**	5.0 ▲	0.00	4.0	4.0 ▽	1.75
rxr_2018_n49	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
rxr_2019_n49	5	**3.0**	3.0 ▲	0.75	2.0	2.0 ▽	0.00
rxr_2020_n49	5	**5.0**	5.0	0.00	**5.0**	5.0	0.00
rxr_2021_n49	5	**5.0**	4.5 ▽	1.00	**5.0**	5.0 ▲	0.00
rxr_2022_n49	5	**4.99**	4.99	0.00	**4.99**	4.99	0.00

Table 29: Results on MS-LOP for Δ_{SP} metric and $m = 10$ on RXR instances

Instance	**m**	**MS-CD-RVNS**			**MS-MA-EDM**		
		Best	**Median**	**IQR**	**Best**	**Median**	**IQR**
rxr_1995_n49	10	**9.97**	9.96	0.00	9.96	9.96	0.00
rxr_1996_n49	10	5.0	5.0 ▽	1.00	**7.0**	7.0 ▲	1.00
rxr_1997_n49	10	**9.4**	9.1 ▲	0.90	8.5	8.5 ▽	0.30
rxr_1998_n49	10	**8.0**	8.0 ▲	1.00	5.0	6.0 ▽	1.00
rxr_1999_n49	10	**9.92**	9.92 ▲	0.01	9.9	9.9 ▽	0.00
rxr_2000_n49	10	**9.07**	9.07	0.00	**9.07**	9.07	0.30
rxr_2001_n49	10	**9.09**	9.09	0.00	**9.09**	9.09	0.00
rxr_2002_n49	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
rxr_2003_n49	10	**8.97**	6.98 ▲	0.99	5.98	5.98 ▽	1.00
rxr_2004_n49	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
rxr_2005_n49	10	**9.99**	9.99	0.00	**9.99**	9.98	0.01
rxr_2006_n49	10	5.7	6.7 ▽	1.30	**8.4**	7.75 ▲	1.00
rxr_2007_n49	10	**7.0**	6.0 ▲	1.00	6.0	5.0 ▽	1.00
rxr_2008_n49	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
rxr_2009_n49	10	**9.09**	8.79 ▲	0.60	8.49	8.49 ▽	0.00
rxr_2010_n49	10	**9.0**	9.0 ▲	1.00	7.0	8.0 ▽	1.00
rxr_2011_n49	10	5.0	5.0 ▽	1.00	**6.0**	6.0 ▲	0.00
rxr_2012_n49	10	**6.0**	6.0	1.00	5.0	6.0	1.00
rxr_2013_n49	10	**10.0**	10.0	0.00	9.0	10.0	1.00
rxr_2014_n49	10	6.0	7.0 ▽	1.00	**10.0**	10.0 ▲	0.00
rxr_2015_n49	10	9.95	9.96 ▲	0.74	**9.97**	7.97 ▽	2.00
rxr_2016_n49	10	7.7	7.7 ▽	1.62	**8.8**	8.8 ▲	0.00
rxr_2017_n49	10	**9.0**	9.0 ▲	1.00	4.0	4.0 ▽	1.75
rxr_2018_n49	10	**10.0**	10.0 ▲	0.23	9.7	9.7 ▽	0.53
rxr_2019_n49	10	**3.0**	3.0 ▲	0.75	2.0	2.0 ▽	0.00
rxr_2020_n49	10	**10.0**	10.0	0.00	**10.0**	10.0	0.00
rxr_2021_n49	10	5.0	4.5 ▽	1.00	**6.0**	5.0 ▲	1.00
rxr_2022_n49	10	**9.98**	9.98	0.00	**9.98**	9.98	0.00

Table 30: Results on MS-LOP for Δ_{SP} metric and $m = 15$ on RXR instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
rxr_1995_n49	15	**14.95**	14.95 ▲	0.01	14.94	14.94 ▽	0.00
rxr_1996_n49	15	5.0	5.0 ▽	1.00	**7.0**	7.0 ▲	1.00
rxr_1997_n49	15	**14.4**	13.5 ▲	0.50	12.9	12.9 ▽	0.00
rxr_1998_n49	15	**8.0**	8.0 ▲	1.00	5.0	6.0 ▽	1.00
rxr_1999_n49	15	**14.9**	14.91 ▲	0.02	14.89	14.89 ▽	0.01
rxr_2000_n49	15	**14.04**	13.74 ▲	0.30	13.74	13.45 ▽	0.29
rxr_2001_n49	15	**14.09**	14.09 ▲	0.00	9.09	12.09 ▽	2.00
rxr_2002_n49	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
rxr_2003_n49	15	**8.97**	6.98 ▲	0.99	5.98	5.98 ▽	1.00
rxr_2004_n49	15	**15.0**	15.0 ▲	0.00	**15.0**	14.5 ▽	2.00
rxr_2005_n49	15	**14.99**	14.99 ▲	0.01	11.99	12.48 ▽	2.00
rxr_2006_n49	15	5.7	6.7 ▽	1.30	**8.4**	7.75 ▲	1.00
rxr_2007_n49	15	**7.0**	6.0 ▲	1.00	6.0	5.0 ▽	1.00
rxr_2008_n49	15	**15.0**	15.0	0.00	**15.0**	15.0	0.00
rxr_2009_n49	15	**13.49**	13.19 ▲	1.13	12.89	12.89 ▽	0.00
rxr_2010_n49	15	**9.0**	9.0 ▲	1.00	7.0	8.0 ▽	1.00
rxr_2011_n49	15	5.0	5.0 ▽	1.00	**6.0**	6.0 ▲	0.00
rxr_2012_n49	15	**6.0**	6.0	1.00	5.0	6.0	1.00
rxr_2013_n49	15	**13.0**	13.0 ▲	1.00	9.0	10.0 ▽	1.00
rxr_2014_n49	15	6.0	7.0 ▽	1.00	**11.0**	10.0 ▲	1.00
rxr_2015_n49	15	**11.95**	9.97 ▲	2.73	9.97	7.97 ▽	2.00
rxr_2016_n49	15	7.7	7.7 ▽	1.62	**13.2**	12.9 ▲	1.70
rxr_2017_n49	15	**9.0**	9.0 ▲	1.00	4.0	4.0 ▽	1.75
rxr_2018_n49	15	**15.0**	14.7 ▲	0.30	12.7	11.35 ▽	2.93
rxr_2019_n49	15	**3.0**	3.0 ▲	0.75	2.0	2.0 ▽	0.00
rxr_2020_n49	15	14.4	14.4 ▽	0.48	**15.0**	15.0 ▲	0.00
rxr_2021_n49	15	5.0	4.5 ▽	1.00	**6.0**	5.0 ▲	1.00
rxr_2022_n49	15	**14.98**	14.98 ▲	0.00	9.98	11.98 ▽	1.99

Figure 1: Φ vs.Δ_{NN} for all the os300 instances with $m = 5$.

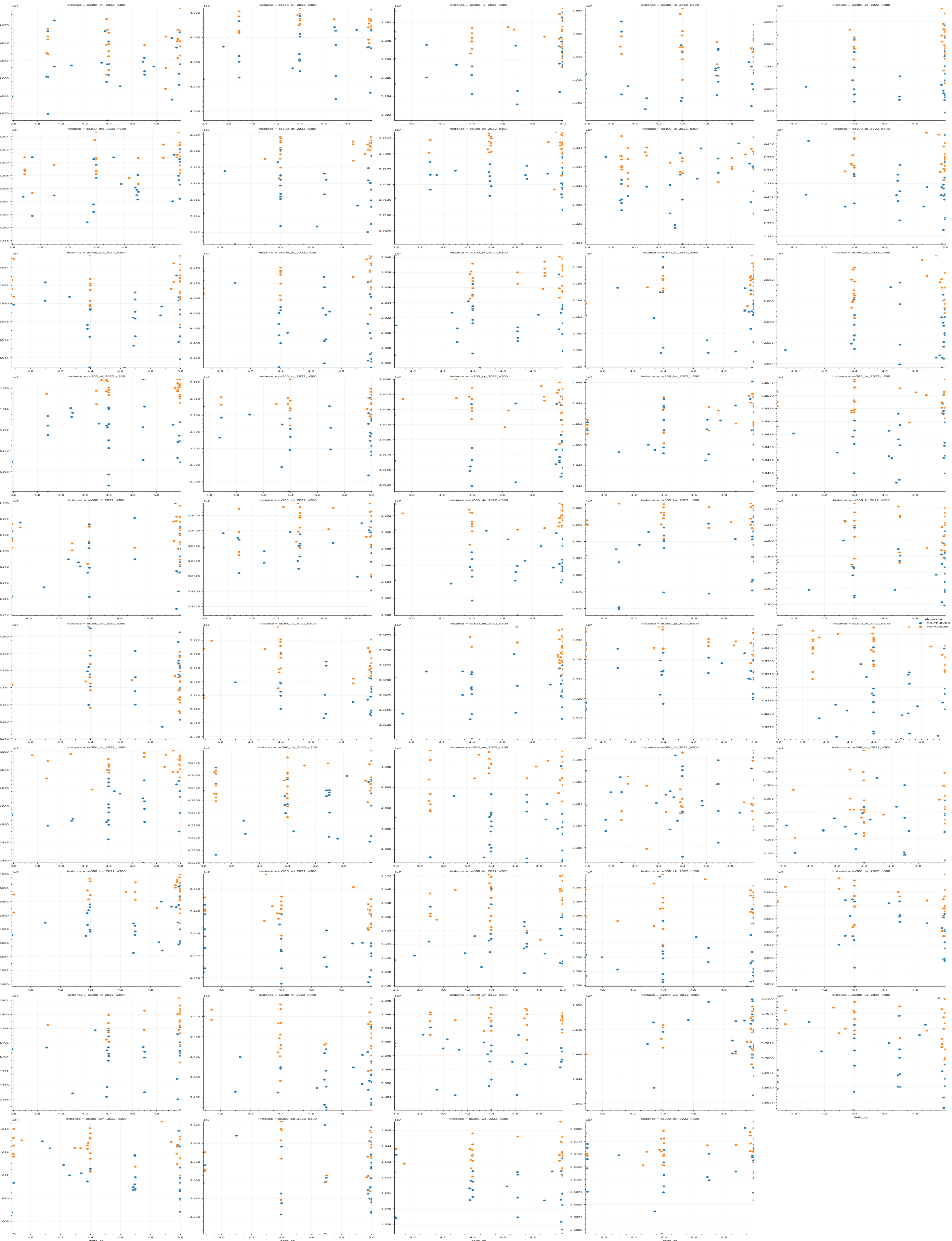

Figure 2: Φ vs.Δ_{SP} for all the os300 instances with $m = 5$.

Figure 3: Φ vs.Δ_{NN} for all the os300 instances with $m = 10$.

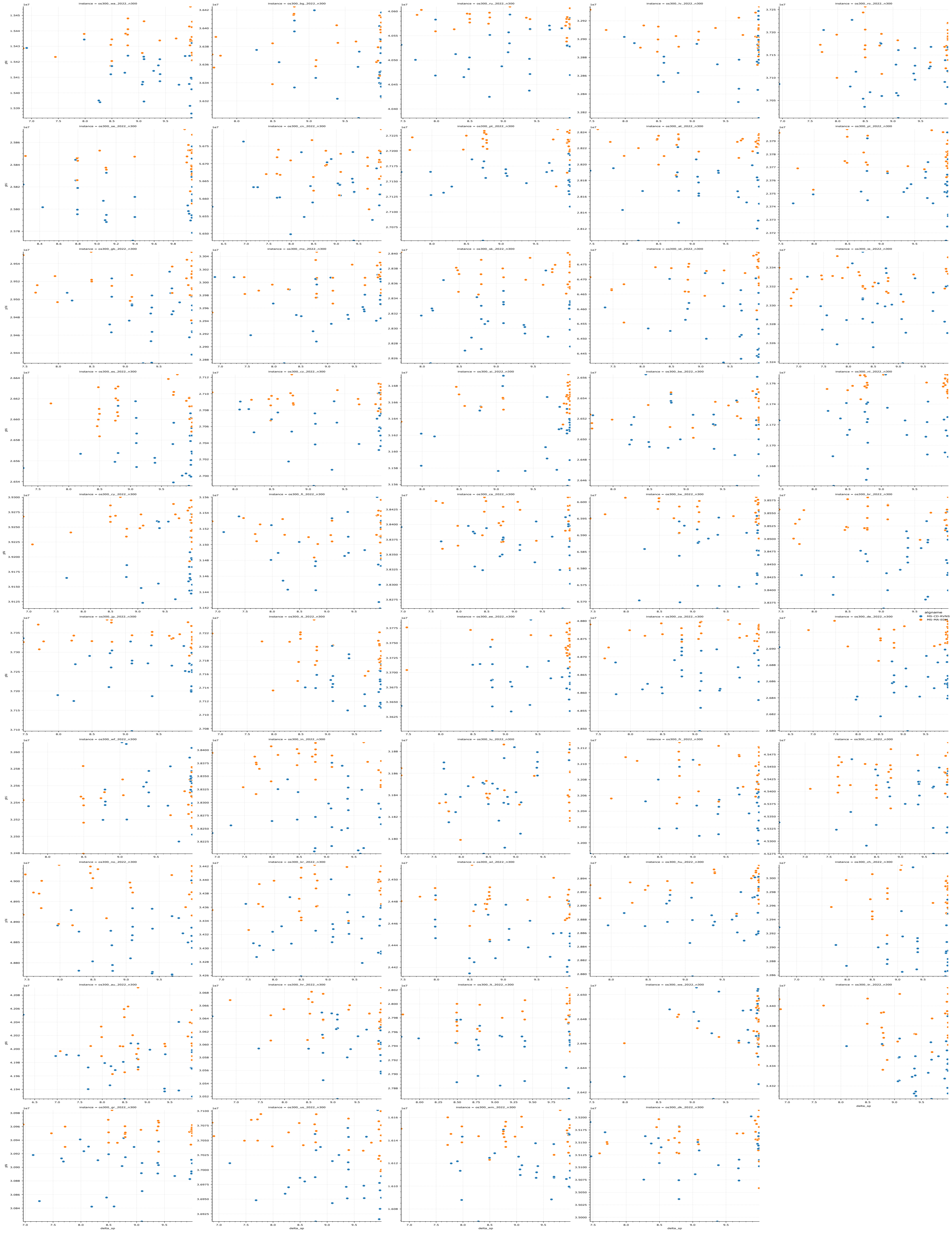

Figure 4: Φ vs.Δ_{SP} for all the os300 instances with $m = 10$.

Figure 5: Φ vs.Δ_{NN} for all the os300 instances with $m = 15$.

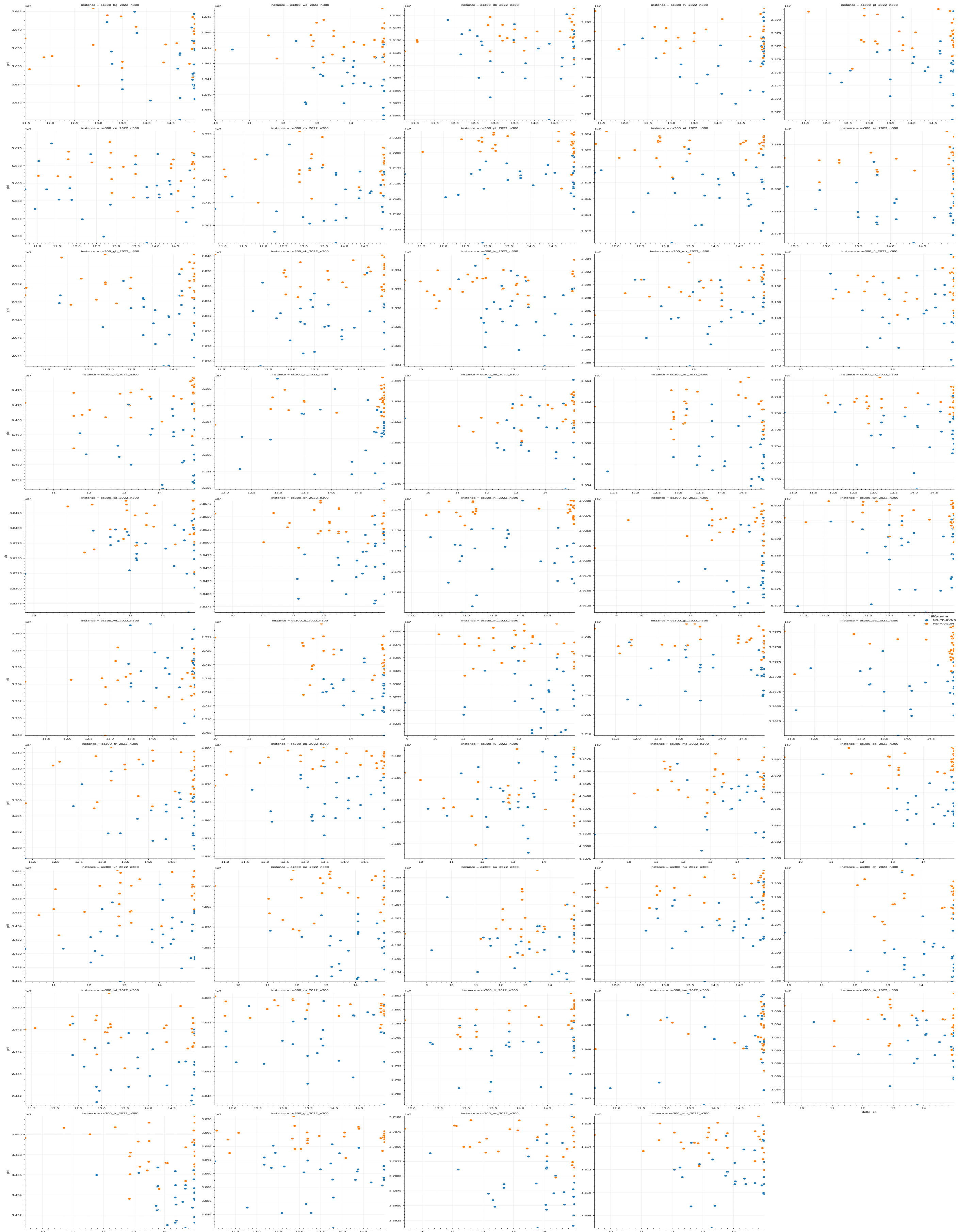

Figure 6: Φ vs.Δ_{SP} for all the os300 instances with $m = 15$.

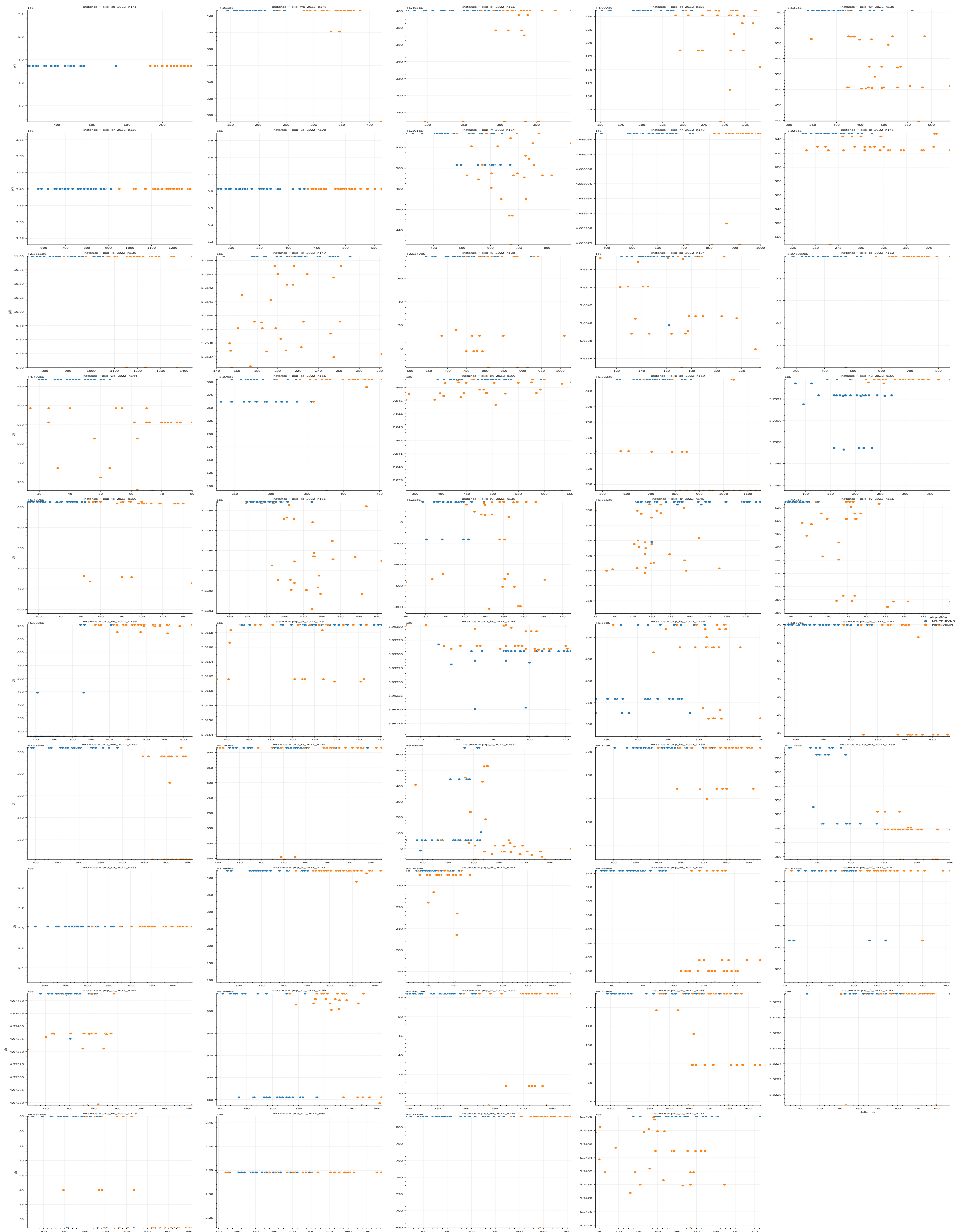

Figure 7: Φ vs.Δ_{NN} for all the pxp instances with $m = 5$.

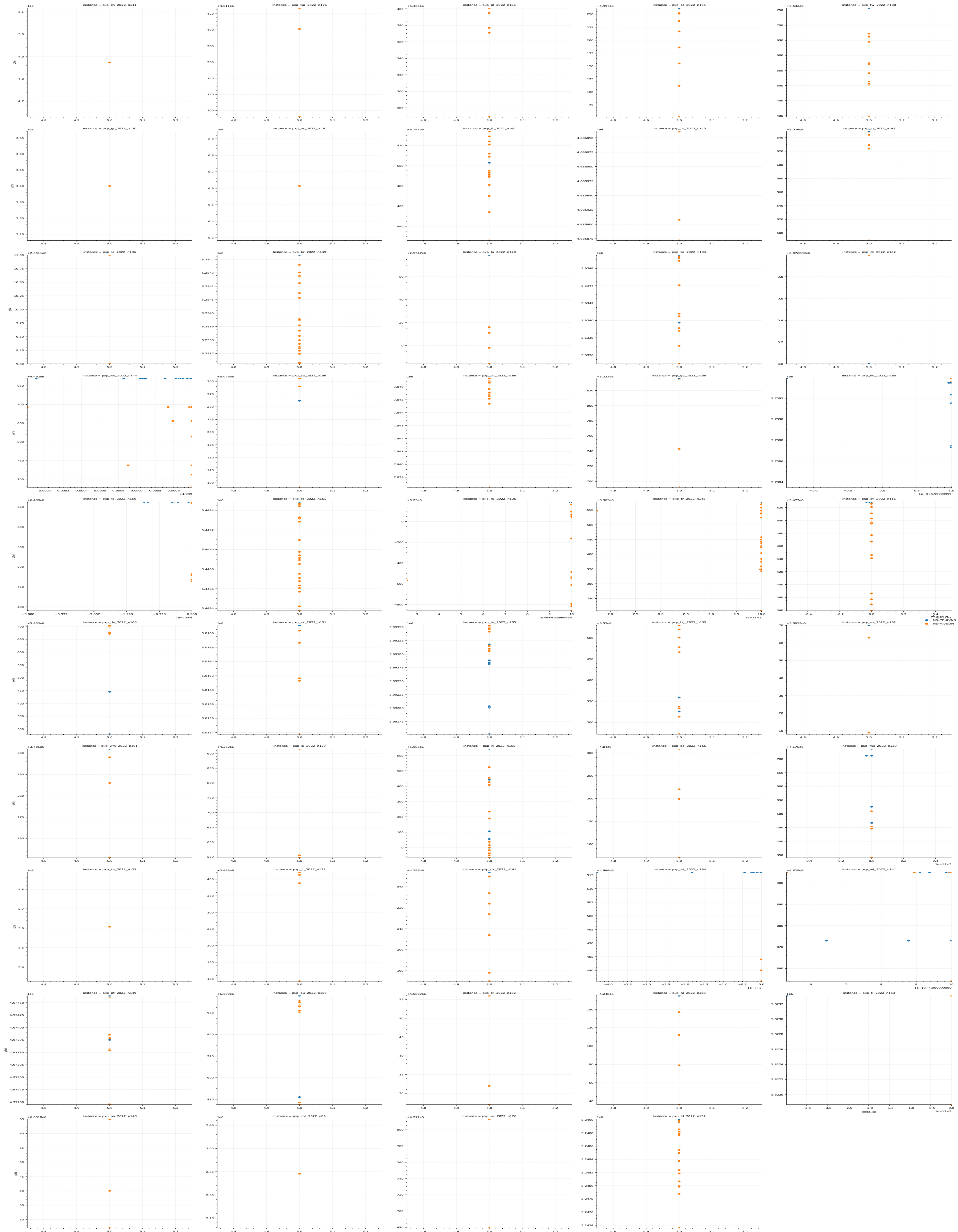

Figure 8: Φ vs.Δ_{SP} for all the pxp instances with $m = 5$.

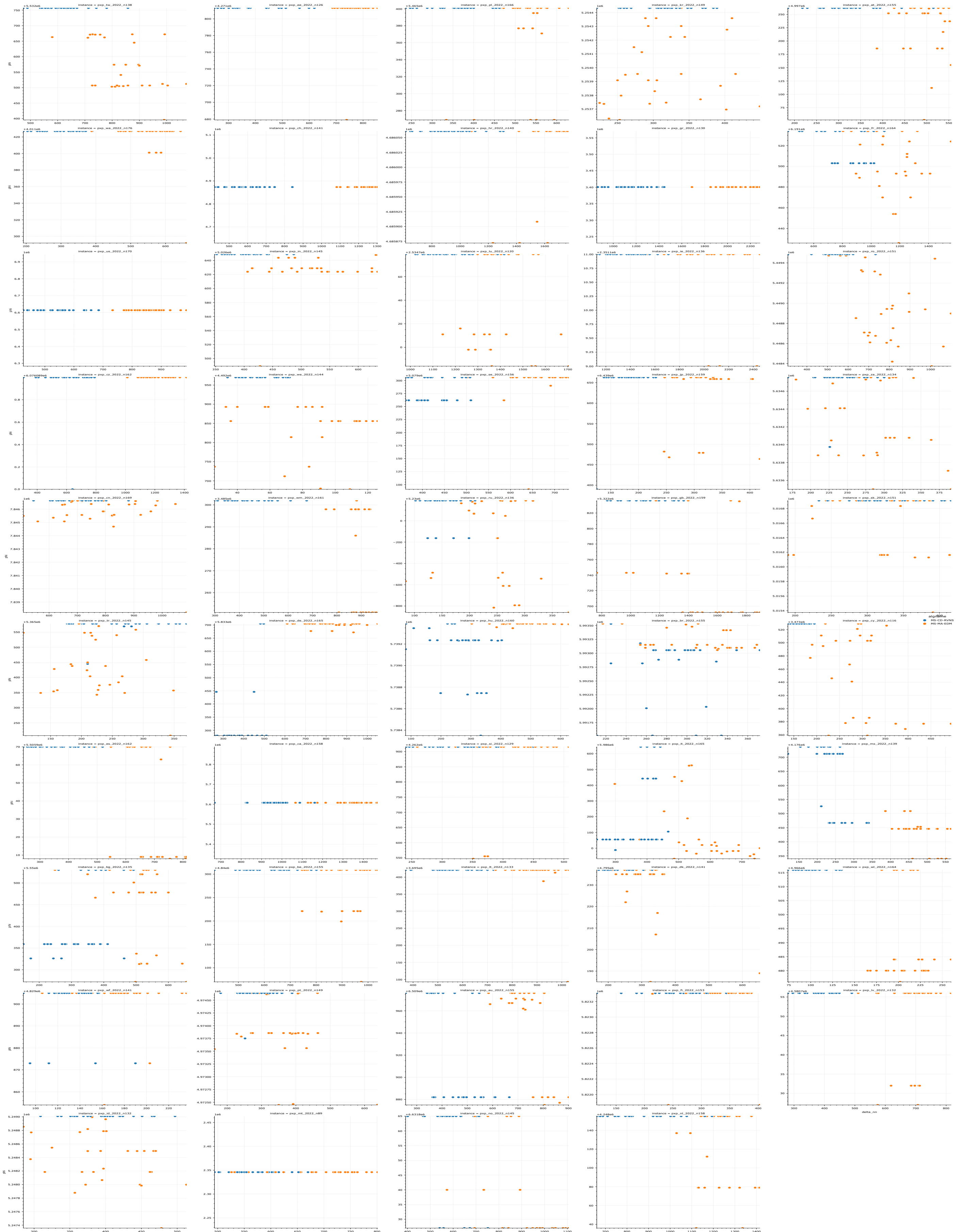

Figure 9: Φ vs.Δ_{NN} for all the pxp instances with $m = 10$.

Figure 10: Φ vs.Δ_{SP} for all the pxp instances with $m = 10$.

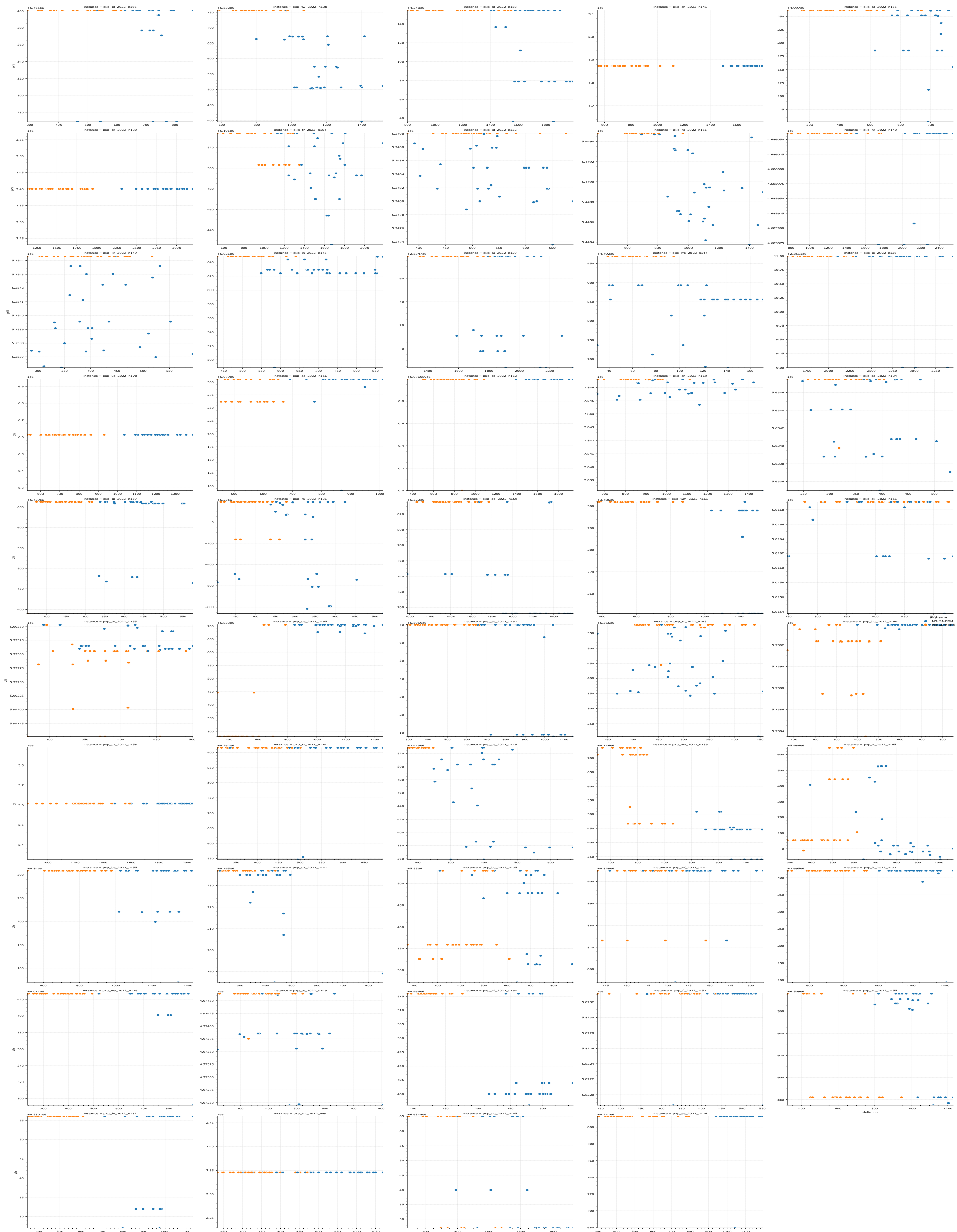

Figure 11: Φ vs. Δ_{NN} for all the pxp instances with $m = 15$.

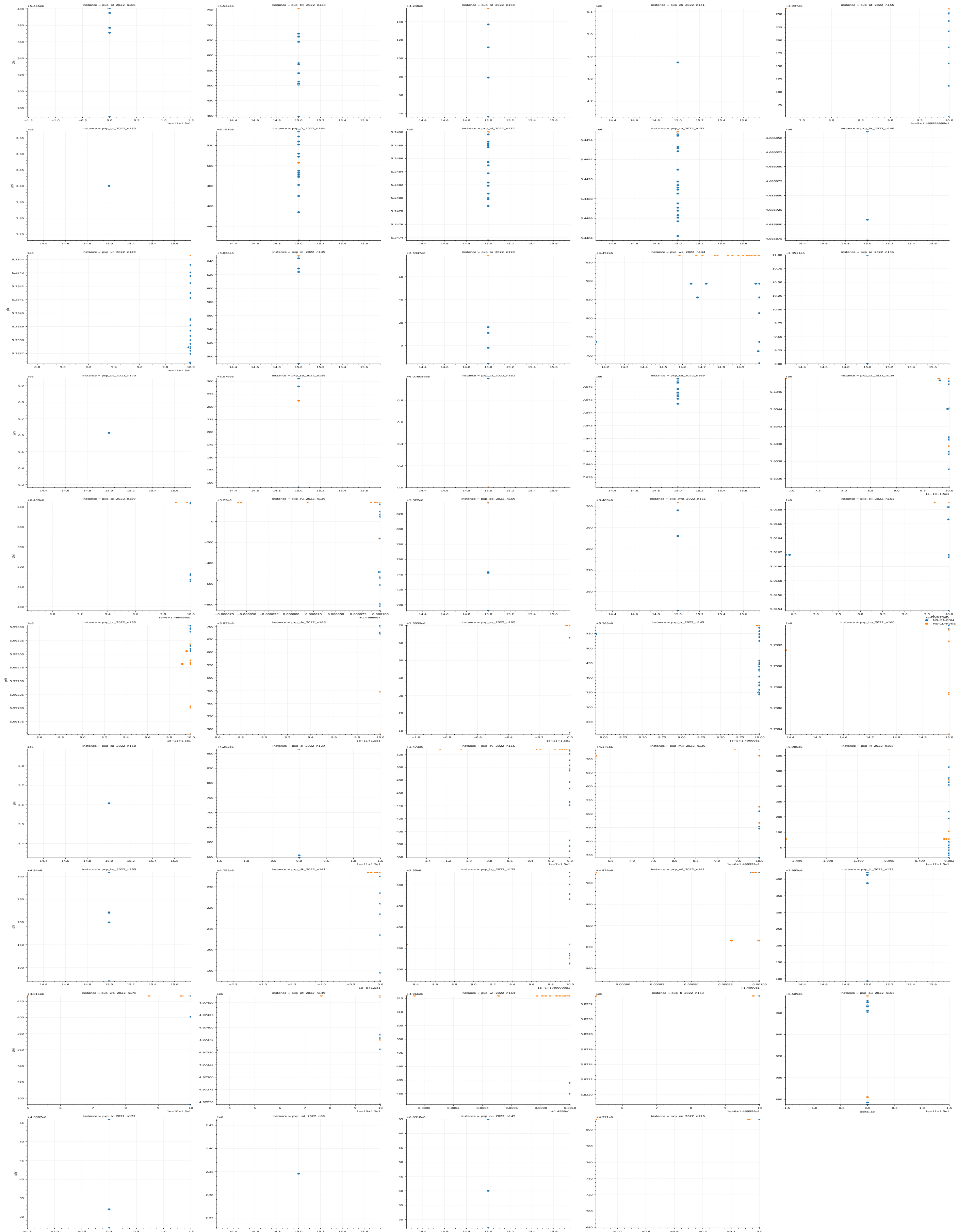

Figure 12: Φ vs.Δ_{SP} for all the pxp instances with $m = 15$.

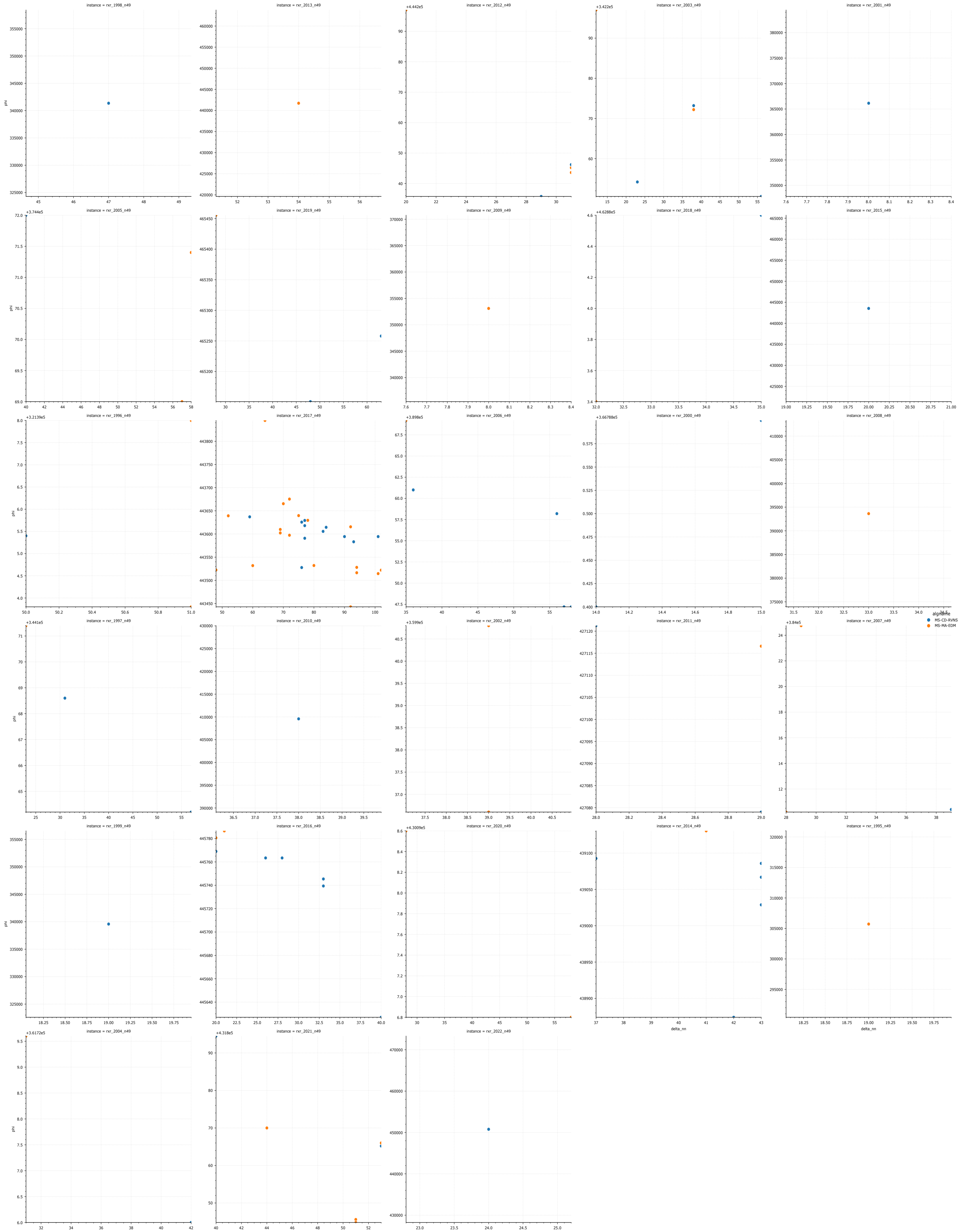

Figure 13: Φ vs.Δ_{NN} for all the rxr instances with $m = 5$.

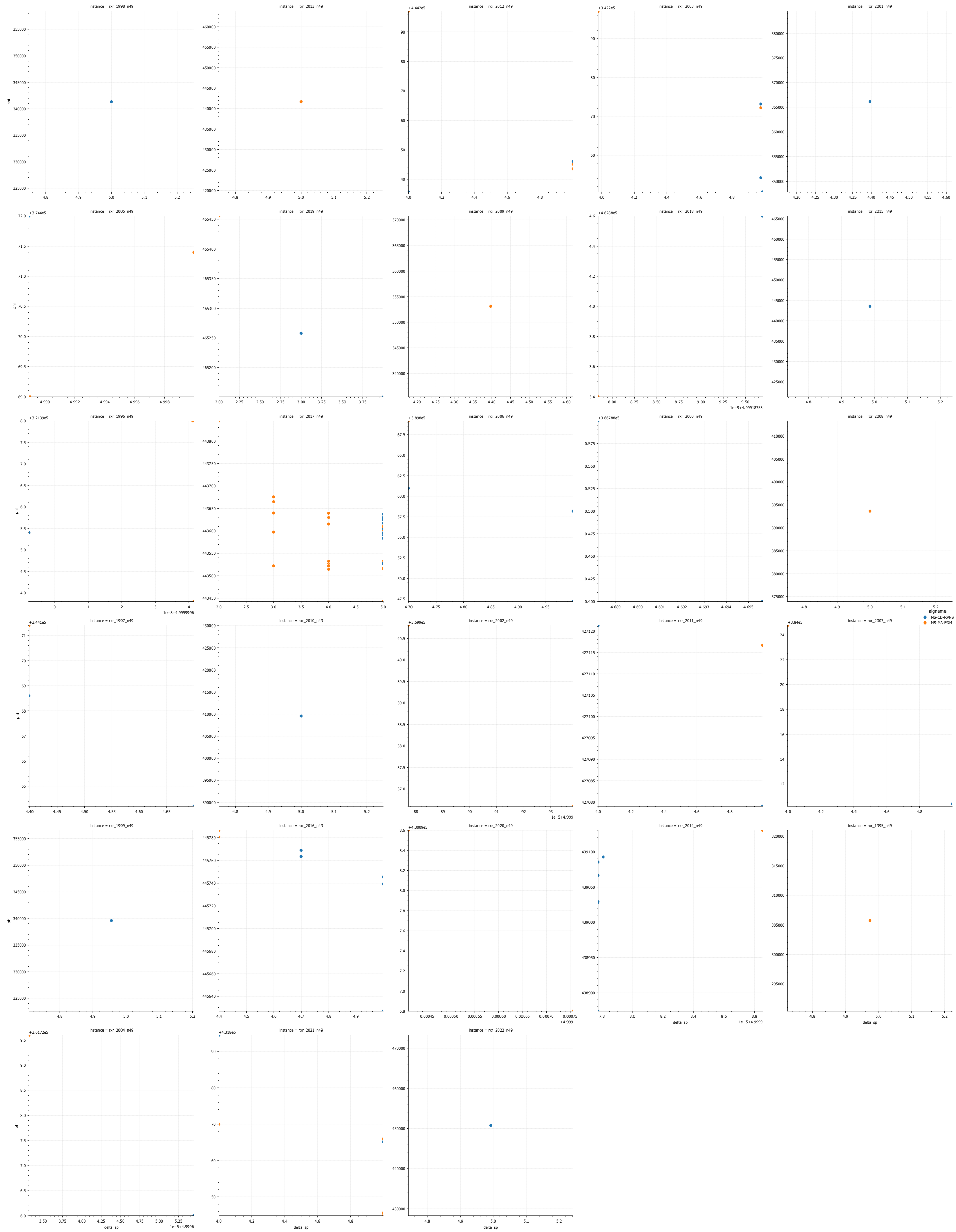

Figure 14: Φ vs.Δ_{SP} for all the rxr instances with $m = 5$.

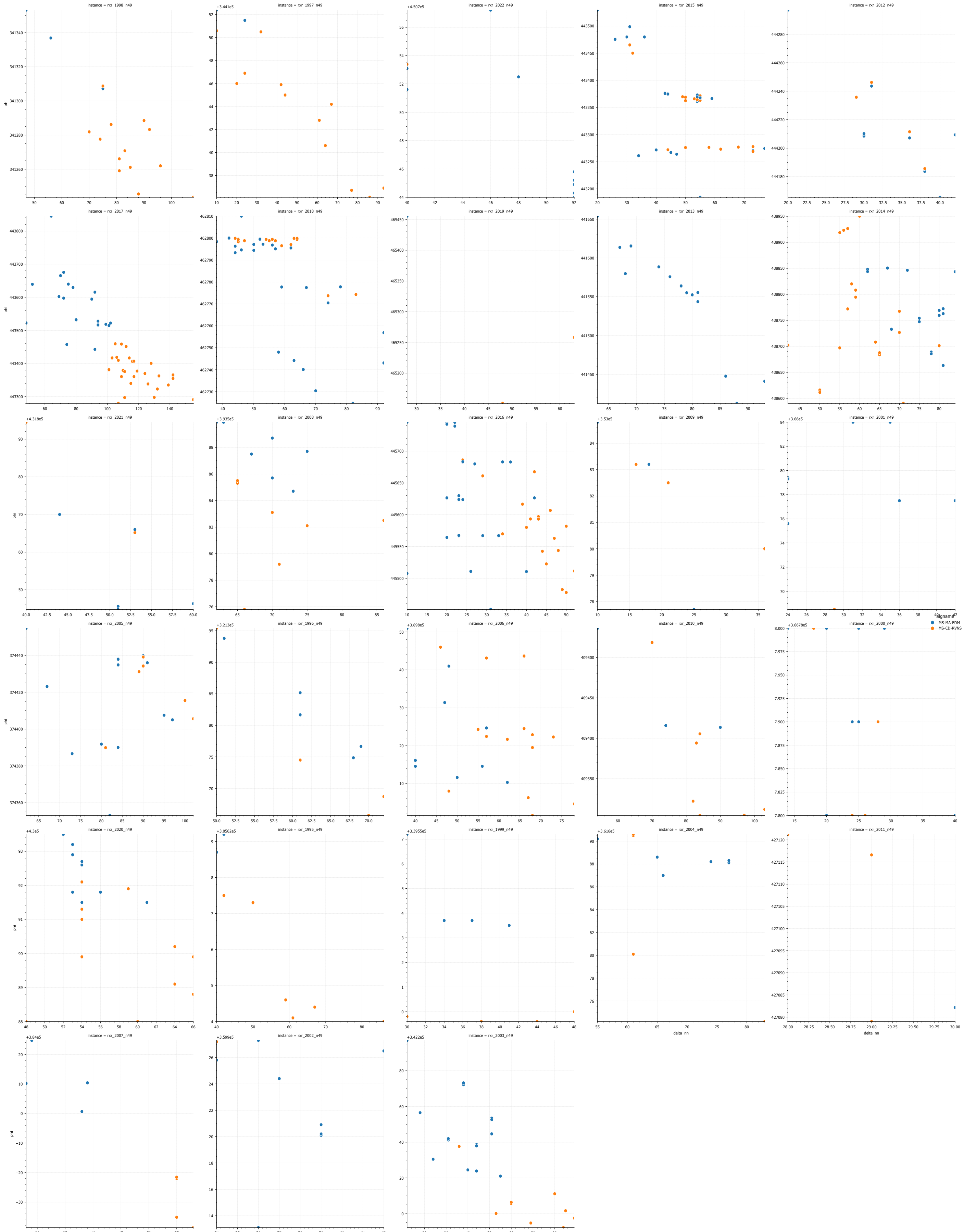

Figure 15: Φ vs.Δ_{NN} for all the rxr instances with $m = 10$.

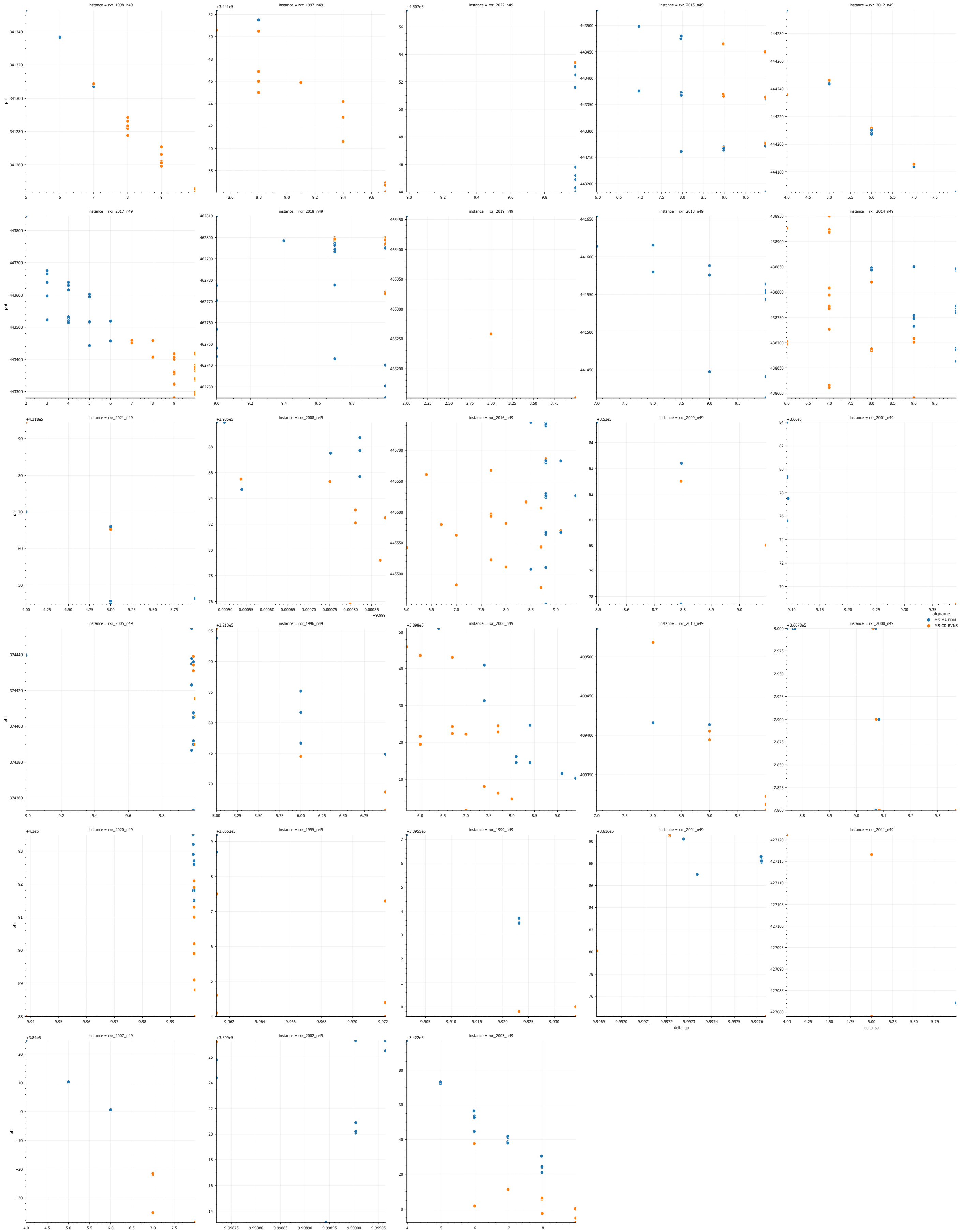

Figure 16: Φ vs.Δ_{SP} for all the rxr instances with $m = 10$.

Figure 17: Φ vs.Δ_{NN} for all the rxr instances with $m = 15$.

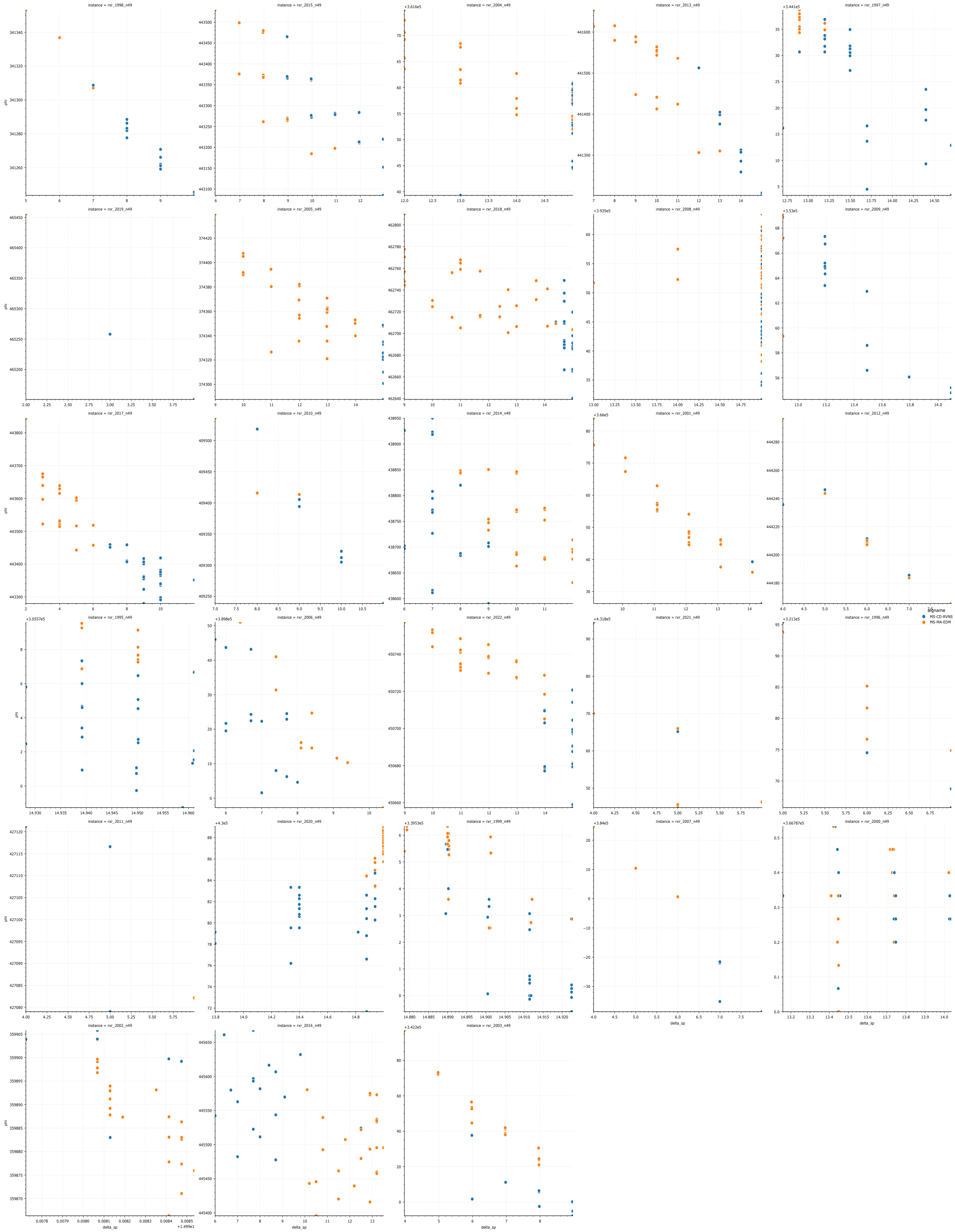

Figure 18: Φ vs.Δ_{SP} for all the rxr instances with $m = 15$.